

Swiss Finance Institute

Research Paper Series

N°24-85

Relationship Banking: The Borrower's
Incentives Channel



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A B S T R A C T

What is the impact of a prior banking relationship on a borrower's incentives to avoid default? To answer this question, we exploit a unique institutional setting and data on 149,230 mortgage loans tracked for twenty four months. We find that the presence of a relationship diminishes borrower's default risk by approximately 4%, attributable exclusively to the value of the relationship for the borrower. This effect persists during the notable surge in loan defaults during COVID-19 and is even more pronounced for borrowers affected by a natural disaster, as well as among wealthier and married male borrowers.

Keywords: Relationship Banking, Borrower's Incentives, Mortgage Loan, COVID-19, Default Risk.

JEL Classification: G20, G21

1. Introduction

Relationship banking and its role in default behavior are widely recognized to operate through three primary channels: screening, monitoring, and borrower’s incentives (Puri et al., 2017). The motivation behind the borrower’s incentives channel stems from the borrower’s preference for preserving the value of the relationship they have built with their bank over time. The relationship helps clients gain easier access to bank loans, often with better terms (Petersen and Rajan, 1994; Berlin and Mester, 1999; Boot, 2000; Puri et al., 2017 among others). Moreover, Bodenhorn and Xing (2025) show that long-term banking relationships not only enable borrowers to obtain larger loans but also lower borrowing costs and collateral requirements.

While screening and monitoring mechanisms have been extensively studied (e.g., Diamond, 1984; Manove et al., 2001; Keys et al., 2010; Puri et al., 2017; Bedayo et al., 2020), the borrower’s incentives channel remains relatively underexplored due to the challenges in distinguishing its impact from that of other channels. To address this challenge, Schoar (2012) for example conducts a firm-level randomized experiment where loan officers are triggered to increase personal interactions with a random group of borrowers (the treatment group). The study demonstrates that borrowers with more personal interactions with loan officers exhibit stronger incentives to avoid default.¹

Our paper aims to complement this experimental approach by elucidating the extent to which a prior relationship with a bank in a unique real world setting influences a borrower’s incentives to avoid default. We focus on mortgage loans and differentiate between borrowers with a prior relationship with the lending bank and those without. In this way we capture *all* factors playing a role in the development of a bank-borrower relationship including actual repeated personal interactions.

The unique setting we study is one where all borrowers — with and without a prior relationship — undergo *identical* screening and monitoring processes, thereby enabling us to *isolate* the impact of borrower’s incentives on loan default from these other two processes.

The setting consists of a singular specialized bank for mortgage loans in Iran, known as “Maskan Bank”. This bank issues so-called “mortgage rights” in the stock market. Applicants *must purchase* these mortgage rights in the market to be eligible for mortgage loans,² and the bank *is obliged* to grant credits to the holders of these mortgage rights. In addition, all holders of mortgage rights, independent of having

¹ Puri et al. (2017) use the activity score — a comprehensive measure of a borrower’s checking account activity — to assess a borrower’s incentives to avoid default. This score includes factors such as the number of checking accounts, account balances, credit line usage, credit limit violations, and bounced checks, weighted for their predictive power of defaults. However, their study neither distinguishes borrowers with a bank relationship from those without one, nor does it fully separate the borrower’s incentives channel from the screening and monitoring channels.

² The loan purposes encompass various activities related to housing, including purchasing, construction, renovation, and repairing a house.

any prior relationships with Maskan Bank, are treated similarly in the loan screening and monitoring processes. This unique setting ensures that any difference in a borrower's repayment behavior is attributable solely to factors beyond monitoring and screening, i.e., the borrower's incentives channel.

To illustrate the mechanism, we develop a conceptual model in Appendix A1. The model begins with an intertemporal consumption-smoothing framework that formalizes how borrowers weigh the long-term benefits of maintaining a banking relationship — such as continued access to credit and more favorable loan terms — against the short-term gains from default. In doing so, it provides a theoretical basis for understanding default behavior as a function of borrowers' incentives to preserve their relationship with the bank, which effectively serves as reputational collateral.

We then turn to a comprehensive dataset on 149,230 mortgage loans and their monthly performance records from March 2019 to February 2021, resulting in around 1,500,000 loan-month observations.³ This dataset offers detailed insights into both loans and borrowers' characteristics, including information on pre-existing relationships between borrowers and the bank.

Our analysis shows that the borrowers with prior relationships exhibit approximately a 4% lower risk of default compared to those without such relationships. The findings underscore the importance of a borrower's incentives to avoid default and, consequently, maintain the relationship developed with the bank. This is consistent with [Bodenhorn and Xing \(2025\)](#), who show that relationship borrowers, who default, experience a deterioration in their banking relationships capital, resulting in smaller loans and stricter lending conditions. Consequently, borrowers with established relationships have strong incentives to maintain them in order to preserve the benefits of long-term banking relationships. Furthermore, borrowers who lack a prior relationship with Maskan Bank generally maintain relationships with other commercial banks, making default on Maskan Bank's loans less consequential for them. This borrower strategy aligns with the literature, which suggests that borrowers may prioritize protecting relationships with their banks. As a result, they might selectively choose which obligations to fulfil and which bank to default on ([Schäfer, 2019](#)).

We also examine the correspondence during the COVID-19 pandemic and catastrophic events, specifically focusing on the 2020 earthquake in Iran's West Azerbaijan and East Azerbaijan provinces. The analysis reveals that borrowers with pre-existing relationships experienced significantly lower default rates during the COVID-19 pandemic. Similarly, these borrowers were more likely to pay their loan installments after the earthquake, with notably lower default rates observed in the month following the disaster. Overall, the findings highlight the importance of relationship banking in times of crisis and hardship.

³ While the sample includes 149,230 mortgage loans and the observation window spans 24 months, not all loans are observed for the full duration, as some enter the dataset at different times based on their origination dates.

Next, we examine whether the value of relationship and the incentive to maintain it varies across different clientele. Specifically, we investigate the role of wealth, gender, and marital status. On the one hand, it can be argued that banking relationship is more valuable for wealthier borrowers as they are more likely to re-use it in the future. On the other hand, these borrowers might assign less value to relationship banking given that they might be able to post better hard information. Therefore, we investigate whether the value of relationship banking diminishes with wealth. The result of our analysis shows that wealth strengthens the relationship between prior banking relationships and loan performance suggesting that the value of a banking relationship actually increases with wealth.

Since women face discrimination in access to credit ([Bellucci et al., 2010](#); [Alesina et al., 2013](#); [Ongena and Popov, 2016](#)), prior relationships can provide them greater access to loans compared to men ([Abedifar et al., 2024](#)). Consequently, one might expect banking relationships to be more valuable for female borrowers than male borrowers. However, given that men are often more likely to engage in future business dealings, they might also benefit significantly from strong banking relationships. To investigate this, we examine the intermediary effect of gender on the relationship between prior banking relationships and loan performance.

Moreover, the potential benefits of banking relationships might depend on clients' marital status. It can be argued that banking relationships are more valuable for married borrowers because they often demonstrate greater commitments to their professions and responsibilities. Hence, one might expect them to have a stronger inclination to maintain prior banking relationships and fulfill their financial commitments more effectively than single borrowers. Our findings indicate that the relationship effect is particularly pronounced among married male borrowers. Specifically, married men with prior banking relationships exhibit significantly stronger incentives to fulfill their mortgage payment obligations compared to other borrower groups (single women, single men, and married women), thereby reinforcing the importance of relationship value in shaping repayment behavior.

This study contributes to the existing literature on relationship banking in several ways. While the contract model of [Boot and Thakor \(1994\)](#) and the reputation model of [Shapiro \(1983\)](#) implicitly suggest that borrowers invest in their reputation to enhance their ability to smooth consumption over time, neither explicitly models the consumption-smoothing mechanism within the context of bank-borrower relationships. In contrast, our study extends these frameworks by introducing an intertemporal consumption-smoothing model that formally captures how borrowers' incentives to repay a loan are shaped by the strength of their relationship with the bank.

Moreover, our unique institutional setting enables us to investigate the role of the borrower's incentives in loan repayment using a sample of borrowers who are subject to similar screening and monitoring processes. Our analysis provides strong evidence in support of the incentive channel brought up by [Puri](#)

et al. (2017). We also extend [Schoar's \(2012\)](#) research on the role of personal interaction of loan officers in loan performance by emphasizing the importance of prior banking relationships.

The paper also aligns with the strand of literature emphasizing banks' incentives in sustain relationships with preferred clients. For example, [Bolton et al. \(2016\)](#) show that relationship banks in Italy offered favorable terms to profitable borrowers after the failure of Lehman Brothers. Similarly, [Bodenhorn and Xing \(2025\)](#) find that banks extend larger loans on more favorable terms – lower interest rates, fewer endorsers, and with reduced collateral requirements – to relationship borrowers. Our contribution, however, shifts the focus to the incentives of borrowers. We provide evidence that borrowers have strong incentives to preserve their relationships: access to favorable credit terms and future credit encourage repayment. This borrower-centered perspective not only complements the literature on banks' behaviors but also helps explain why banks act as they do.

This paper is also related to the important policy debate about the role of branch banking in the economy ([Burgess and Pande, 2005](#); [Agarwal and Hauswald, 2010](#); [Bruhn and Love, 2014](#); [Nguyen, 2019](#); [Levine et al., 2020](#)). While banks have been shutting down branches at an unprecedented pace, extensive researches demonstrate that branches can influence borrowers' performance and behavior ([Garmaise and Moskowitz, 2006](#); [Bonfim et al., 2023](#); [Benincasa, 2024](#)), which our study also provides robust evidence reinforcing this perspective. Additionally, we shed light on the role of banking relationship during the COVID-19 pandemic, which offers new insights into the dynamics of borrowers' behavior during an economic turmoil. Finally, we demonstrate that the value of banking relationship varies across different customer segments. In particular, our findings suggest that banking relationship is more valuable for wealthier and male borrowers.

The results of this study suggest that banks can increase borrowers' incentives to avoid default by enhancing the value of relationship banking. This can be achieved by improving the reusability of relationship banking and emphasizing its long-term benefits. For instance, banks can reward borrowers with strong repayment histories by offering better loan terms, tiered loyalty benefits, and priority access to other financial services. These policies provide borrowers with greater incentives to maintain a positive track record with the banks. This aligns with [Padilla and Pagano \(1997\)](#), who argue that banks can encourage borrowers to exert sufficient effort to properly implement their projects by committing to share credit information.

The remainder of the paper is structured as follows: Section 2 outlines the unique institutional setting of our study. In Section 3, we detail the methodology employed to assess the impact of the relationship on borrowers' incentives. Section 4 elucidates our data collection process and summary statistics. The findings of our analysis are presented in Section 5. To ensure the robustness of our findings, Section 6 presents a series of robustness checks. Section 7 provides further analysis of our study. Finally, Section 8 offers concluding remarks.

2. Institutional Setting

In this study, we exploit a unique mortgage market setting in Iran that provides an ideal platform to examine the role of borrowers' incentives in loan performance. "Maskan Bank" (commonly referred to as the Housing Bank) is the country's primary provider of housing finance and plays a central role in implementing national housing policy. As a government bank, it extends credit to retail homebuyers through two main programs, while also functioning as a full-service commercial bank offering the same retail banking services as other Iranian banks (see Appendix A2 for a comparative overview of mortgage characteristics in the U.S., Europe, and Maskan Bank programs).

First Program. Prospective first-time homebuyers must deposit half of the loan amount into a Maskan Bank account for one year, without interest, as a prerequisite for receiving a mortgage at an official 8% interest rate.⁴ Because this scheme does not allow us to study the role of borrower's incentives in loan repayment, it lies outside the scope of this paper.

Second Program. On the 15th of each month, the bank issues between 1 and 4 million tradeable "mortgage rights," with issuance volume linked to the bank's aggregate savings deposits. These rights, valid for two years, are traded on the market through a limit order book system. Applicants seeking mortgage loans under this scheme must purchase a certain number of rights and submit them to Maskan Bank.⁵ Unlike the first program, this scheme is not restricted to first-time homebuyers.

A key feature of this program is that Maskan Bank must issue mortgages to any applicant who acquires the required rights and meets the credit score threshold, *regardless* of whether they have prior relationships with the bank. Credit risk assessment is conducted through a centralized system designed and overseen by the central bank, which classify applicants into categories ranging from E3 (highest risk) to A1 (lowest risk).⁶ Applicants who hold mortgage rights and achieve a rating of C3 or higher are automatically eligible, with no additional screening by Maskan Bank. In addition, loan terms are fully standardized: every borrower pays the same nominal interest rate (17.5%) over the same repayment

⁴ It is worth noting that the average inflation rate during the study period is approximately 30%.

⁵ To illustrate how this mechanism operates, consider that for every 5,000,000 Iranian rials (approximately USD 25 during the study period), borrowers are required to purchase one mortgage right from the market. In our sample, the average market price of a right is USD 2.78. Accordingly, for a loan amounting to the equivalent of USD 10,000, applicants would need to acquire 400 rights, incurring an upfront cost of USD 1,112, about 11% of the loan amount. It is noteworthy that the market price of these rights fluctuates substantially, with a standard deviation of USD 0.75 over the study period, and recorded extremes of USD 1.95 (minimum) and USD 4.44 (maximum). Consequently, although the official mortgage rate remains fixed at 17.5%, the effective rate varies with market demand for rights, ranging between 17.92% and 22.97%, with a historical average of 20.41%. Two additional features of the system are worth noting: a) Holders of mortgage rights may either use them to secure a loan or sell them in the market after a waiting period (typically four months). This restriction curbs speculation and ensures that prices primarily reflect housing loan demand; b) once rights are submitted to obtain a loan, they are voided and lose their market value.

⁶ The credit risk evaluation draws on a range factors, including records of bounced checks, loan defaults, tax arrears, unpaid utility bills, and average account balances across all banks, not merely Maskan Bank.

horizon (144 months), regardless of their credit risk. Furthermore, all borrowers — irrespective of whether they have any prior relationship with the bank — are subject to the same monitoring procedures under this lending program.⁷ Specifically, the bank must contact clients after 90 days of delinquency and initiate legal proceeding to seize collateral after 180 days.

Because screening and monitoring are uniform across all borrowers, regardless of whether they have prior relationships with the bank,⁸ this setting allows us to isolate the impact of such relationships on repayment behavior. We define borrowers with prior relationships as those holding an active Maskan Bank account before applying for the loan, while borrowers without prior relationships are those without such an account, though they may maintain accounts at other banks. The first group has a stronger incentive to honor their repayment commitments: defaulting not only damages their credit history but also erodes their relationship capital with Maskan Bank, which they may rely on for future access to retail banking services.⁹ By contrast, borrowers without prior relationships with Maskan Bank face credit history consequences without jeopardizing the relationship capital that they have built with their primary banks elsewhere. Therefore, any systematic differences in repayment behavior between the two groups reflect the value of maintaining a relationship with Maskan Bank, *ceteris paribus*.¹⁰

A potential concern is that the absence of prescreening may attract systematically riskier borrowers. However, this risk is mitigated by two factors. First, borrowers must pay an upfront fee equal to about 11% of the loan amount when purchasing rights (based on our sample). Second, in our data, the average loan-to-value ratio is just below 35%, indicating that borrowers are relatively middle class, and carry comparatively low leverage.

⁷ By contrast, under conventional lending programs of Maskan Bank — like many other banks — often offers greater flexibility to loyal customers, for instance by providing early warnings before default, granting extensions, allowing temporary forbearance, or rescheduling installments.

⁸ Based on our discussions with a loan officer at a Maskan Bank branch, we confirm that the bank does not differentiate in its treatment of borrowers based on prior relationships. For example, it does not deduct installments from other accounts held by borrowers with prior relationships. In fact, deposits at Maskan Bank, including savings and current accounts, are legally protected and cannot be used to offset unpaid mortgage obligations. This ensures that even borrowers with prior relationships who hold such accounts face no risk of losing their balances, so deposit holdings do not affect their default incentives. The bank also does not provide these borrowers with advance notice of impending defaults. Hence, since the bank's treatment of borrowers is perfectly uniform across those with and without prior relationships, any observed differences in default behavior can be attributed exclusively to borrowers' incentives.

⁹ A related concern is whether defaulting on a mortgage issued by a government-owned bank entails broader or qualitatively different consequences than defaulting on a loan issued by a private bank. This concern is alleviated by the institutional features of the Iranian banking system: loan defaults, whether with government-owned or private lenders, are recorded in a centralized credit registry maintained by the Central Bank and affect borrowers' credit scores in an identical manner. There are no additional administrative or non-financial penalties specific to state-owned banks, and any constraints following default operate exclusively through the credit registry and apply uniformly across all lending institutions. Consequently, the borrower incentives examined in this paper are driven by standard credit-market mechanisms rather than differential enforcement or state-specific treatment.

¹⁰ Subsections 4.2 and 6.2 address concerns about the potential non-randomness in relationship formation.

Finally, it is important to emphasize that borrowers' incentives to maintain banking relationships — driven by improved access to credit and services — are not unique to Iran.¹¹ While Iranian banks often condition lending on relationship histories, similar patterns are well documented internationally (e.g., [Fukuda and Hirota 1996](#) for Japan; [Elsas and Krahnen 1998](#); [Harhoff and Körting 1998](#); [Puri et al. 2017](#) for Germany; [Angelini, Di Salvo, and Ferri 1998](#) for Italy; [Petersen and Rajan 1994](#), [Cole 1998](#), [Berlin and Mester 1999](#), and [Bodenhorn and Xing 2025](#) for the U.S.). Our findings therefore have broader applicability. Hence, although empirical studies often face a trade-off between internal against external validity ([Imbens and Rubin, 2015](#)), in our case external validity is less of a concern, given the institutional parallels in how prior relationships affect access to retail banking services worldwide.¹²

¹¹ We examine whether longer banking relationships are associated with more favorable loan pricing. To do so, we compile a separate dataset of loans issued under the *First Program* of Maskan Bank, as its screening, monitoring, and loan pricing are not uniform across borrowers (unlike the *Second Program*), allowing us to assess whether longer relationships are associated with preferential treatment. Consistent with the relationship-banking literature ([Petersen and Rajan, 1994](#); [Berlin and Mester, 1999](#); [Boot, 2000](#); [Puri et al., 2017](#); [Bodenhorn and Xing, 2025](#)), we find that a one-standard-deviation increase in relationship duration (approximately five years) is associated with a statistically significant (at the 1% level) reduction of 0.275 percentage points in the loan interest rate, relative to an average rate of about 8%. Detailed results are not reported here for brevity but are available from the authors upon request.

¹² One other potential concern regarding the external validity of our findings is that bank-borrower relationships in Iran's residential mortgage market may be unusually strong if mortgage provision is highly concentrated in a single institution. In such a setting, default could severely restrict borrowers' future access to housing finance, amplifying incentives to maintain a good standing with the lender. In this case, our estimation reflects the upper bound of the borrower incentives channel. However, this concern is mitigated by the fact that while Maskan Bank is the largest provider of residential mortgages in Iran, it is not the sole provider. Based on Central Bank of Iran report at the end of 2021, Maskan Bank accounts for approximately 34.5% of total residential mortgage credit, followed by Meli Bank (24.3%), Refah Bank (9.2%), and several other institutions with smaller shares, including Pasargad (7.5%), Mellat (7.1%), and Tejarat (4.5%). The remaining banks each account for less than 3% of the market. Importantly, mortgages issued by Meli Bank under the "Second Program" are institutionally and contractually equivalent to those issued by Maskan Bank: the qualification process, the use of tradable mortgage rights, and the loan terms are effectively identical. In contrast, mortgages issued by other banks, such as Pasargad Bank, follow a structure similar to the "First Program", differing primarily in loan maturity (shorter terms), interest rates, and loan amounts. Hence, although Maskan Bank plays a dominant role, borrowers retain access to alternative lenders and housing-related credit products. Moreover, Maskan Bank operates as a universal bank and provides a broad range of financial services beyond residential mortgages, similar to other Iranian banks. Consequently, borrowers' incentives to preserve their relationship are not limited to maintaining access to future mortgage credit but also extend to other types of borrowing and financial services, including consumer loans, production and business loans, home repair loans, cheque book issuance, and various savings accounts. This broader relationship dimension suggests that the borrower-incentives channel we identify captures general relationship-banking considerations rather than being driven solely by mortgage market dominance. As a result, borrowers who default may jeopardize their access to a wide array of retail banking services in the future, reinforcing the interpretation of our findings.

3. Empirical Model

In this section, we delineate the methodology used to investigate the impact of borrowers' incentives on the probability of default based on a sample of borrowers with a prior relationship with the bank vis-à-vis those without a prior relationship. We estimate the following specification using the Linear Probability Model (LPM), probit, and logit:

$$P(\text{Default})_i = \beta_0 + \beta_1 \text{RelationshipDummy}_i + \beta_2 \text{Characteristics}_i + \beta_3 \text{RiskMeasures}_i + \sum_{t=1}^M \beta_{4t} T_t + \epsilon_i \quad (1)$$

Where i and t subscripts represent borrower and loan origination month, respectively.

In this model, the dependent variable is the borrower's default occurrence within the full observation period after loan origination where default is defined as failure in repaying installments for three consecutive months.¹³ If the borrower records a default during the study period, Default_i takes the value of one, and zero otherwise. On the right-hand side, we categorize independent variables into three distinct sets. The variable of our interest is the prior relationship ($\text{RelationshipDummy}_i$), defined as a dummy variable that takes the value of one if the borrower i holds a savings or current account before obtaining the loan, and zero otherwise. The marginal effect associated with this variable provides a measure of the comparative probability of default between borrowers with a prior relationship and those without such a relationship. To account for potential inherent differences between those with and without a prior relationship, we control for borrower characteristics (Characteristics_i), including gender, marital status, occupation, and educational background. Lastly, we include variables in our model variables that indicate borrower and loan risks (RiskMeasures_i), such as loan-to-value ratio (*Loan to Value*), historical defaults in previous loans by borrowers (*Default in Previous Loans*), records of bounced checks (*Bounced Check*), and mortgage effective rate (*Effective Rate*).

We also include loan origination month fixed effects (T_t) to absorb common macroeconomic shocks and time-varying aggregate conditions — such as inflation, unemployment, and policy responses — that may differ across origination periods. The sample period (March 2019 to February 2021) substantially overlaps with the COVID-19 pandemic. Consequently, loans originated prior to the outbreak were subsequently exposed to the pandemic shock, whereas loans originated later were issued under pandemic conditions.

Additionally, borrowers located in different cities may vary in their engagement with Maskan Bank and in their wealth levels, which could influence default probabilities. To account for these cross-city

¹³ We also use different default horizons of 6-month and 12-month for the dependent variable as a robustness check in subsection 6.3.1.

differences, we include city fixed effects in our specification. The results are reported in column (4) of Table 3.

In addition, we use two-way clustered standard errors at the city and loan origination-month levels. This approach allows for arbitrary correlation in residuals within cities and within origination months, thereby providing more reliable inference.

4. Data and Summary Statistics

We collect proprietary data on mortgage loans, as well as borrowers' characteristics and account features, from Maskan Bank, and compile two datasets.

The mortgage loan dataset is organized as a panel, tracking monthly repayment status of loan contracts issued between March 2019 and February 2021. This dataset includes key loan characteristics such as the mortgage origination date, the price of mortgage right purchased by the borrower, loan amount, collateral value, number of paid installments, and details of the branch such as its name, code, and location (city) of the branch. Since borrowers acquire mortgage rights at different prices, we calculate the effective rate for each loan based on the loan interest rate (17.5%) and the mortgage rights price.

The second dataset comprises borrowers' characteristics and account features. Borrowers' characteristics encompass a variety of attributes, such as gender, marital status, occupation, and education level.¹⁴ In addition to personal information, risk measures are also included in the borrowers' characteristics dataset, which consist of whether the borrower has a recorded default on previous loans or a bounced check record at any bank in Iran. Account features include the type of account,¹⁵ monthly balance, and the dates of account opening and closing. The account opening date allows us to determine whether the borrower has a prior relationship with the bank. Specifically, the relationship indicator is set to one if the borrower has an existing savings or current account prior to applying for the loan, and zero otherwise.

We merge the borrowers' characteristics and account features dataset and the mortgage loan dataset, based on the hash code of borrowers' National IDs presented in both datasets. The final dataset comprises loans and borrowers' characteristics, and performance variables for approximately 180,000 mortgage loans initiated under the first mortgage lending program of Maskan Bank during the period

¹⁴ It is important to note that the occupation and education variables are recorded as numerical codes, with each code corresponding to an occupational group or a specific education level. Since the dataset does not provide the exact definitions of these codes, we control for occupation and education only through their fixed-effects.

¹⁵ Account types in our dataset are classified into two main categories: savings and current accounts.

from March 2019 to February 2021, with the first loan installment status observable in the data in April 2019.

Since we are interested in the effect of a prior relationship on the risk of default, we exclude loans originated during the last two months of the dataset, as there is insufficient time to capture borrowers' defaults. Furthermore, we drop mortgages with inadequate data, e.g., observations with records for only one or two months are excluded. The final sample comprises 149,230 loans. Afterwards, we treat outliers by winsorizing (Chambers et al., 2000) the distribution at the 1% and 99% levels, which trims extreme values on both tails.

4.1. Summary Statistics

The summary statistics of our final dataset are presented in Table 1. For a detailed description of the variables, please refer to Appendix A3.

[Insert Table 1 About Here]

As illustrated in Table 1, the variable related to prior relationships (*RelationshipDummy*) indicates that 65% of loan applicants do not have a savings or current account with Maskan Bank before applying for a mortgage loan. The variable *RelationshipDuration* shows that, among borrowers with prior relationships, the average relationship with Maskan Bank spans 9.33 years.¹⁶ This suggests that these relationships are originally established to access retail banking services rather than to obtain mortgage loans under the *Second Program*. Moreover, the default indicators reveal that around 25% (and 5%) of borrowers experienced at least one recorded default defined as a 3-Month Default (and 6-Month Default), respectively.¹⁷ Notably, the remaining borrowers exited the dataset without exhibiting any defaults.¹⁸

The loan to value ratio (*Loan to Value*) is, on average, 34.56%, ranging between 4.13% and 83.33%. The effective rate of loans (*Effective Rate*) averages 20.42% spanning from 17.91% to 22.97%, reflecting changes in demand for mortgage rights.

Examining the risk indicators in our data, namely the recorded bounced checks of borrowers (*Bounced Check*) and previous loan default records (*Default in Previous Loans*), we find that approximately 3%

¹⁶ Based on the characteristics of the data, we can only use relationship duration as a measure of relationship intensity and cannot incorporate alternative metrics, such as the amount of fees and commissions collected from each borrower as a measure of relationship intensity.

¹⁷ We define “6-Month Default” as a borrower’s failure to repay installments for 6 consecutive months. However, our main dependent variable in our analysis is “3-Month Default”, while we use “6-Month Default” as a robustness test in subsection 6.3.2.

¹⁸ However, it is plausible that defaults occurred post-February 2021. These are known as “right-censored subjects”. We will address right-censored subjects in subsection 6.1.1 using the Cox model by formulating the log-likelihood function as a weighted combination of two components: the sample density of loans experiencing default within the study time and the survivor function representing the censored loans.

of borrowers have at least one bounced check and 12% have at least one recorded default in their previous loan contracts.¹⁹ Lastly, the data shows that approximately 58% of individuals in our dataset are male (*Male*), and 86% of them are married (*Married*). The statistics show that the waiting time to be admitted for a loan after submitting a mortgage application at a branch (*Loan Origination Time*) averages approximately 8 days, with a maximum waiting time of 31 days.

4.2. Covariate Balance

Ensuring well-balanced covariates between borrowers with and without prior relationships is crucial to guarantee that the two groups of borrowers might also be similar in unobservables, and therefore, the results are not spurious. Adopting the methodology suggested by Imbens and Rubin (2015), we utilize three metrics to assess the overlap in covariate distributions:

1) **Normalized Difference:** This measure is defined as the difference in means between the two groups, divided by the square root of the average of the variances of the two samples. Mathematically, it is expressed as: $\Delta_{nr,r} = (\mu_r - \mu_{nr}) / (\sqrt{(\sigma_r^2 + \sigma_{nr}^2)/2})$, where μ , σ , r , and nr denote mean, standard deviation, the sample of borrowers with prior relationship, and the sample of borrowers without prior relationship, respectively.

2) **Dispersion Difference:** This measure is defined as the logarithm of the ratio of the standard deviations of the sample of the borrowers with prior relationships to the sample of the borrowers without prior relationships. It is mathematically represented as: $\Gamma_{nr,r} = \log(\sigma_r / \sigma_{nr})$.

3) **Coverage Frequency:** This measure evaluates the degree of overlap between the two groups by assessing the proportion of borrowers without prior relationships (borrowers with prior relationships) units that have covariate values in the tails of the distribution of that covariate in the borrowers with prior relationship (borrowers without prior relationship) units. It is represented as: $\pi_r^\alpha = (1 - F_r(F_{nr}^{-1}(1 - \alpha/2))) + F_r(F_{nr}^{-1}(\alpha/2))$, where for the specific value of α , π_r^α denotes the probability mass of the covariate's distribution in the sample of the borrowers with prior relationships that is located outside the quantiles $1 - \alpha/2$ and $\alpha/2$ of that covariate in the sample of the borrowers without prior relationships. This measure ensures that the tails of the covariate distribution in one group do not extend beyond the extreme values observed in the opposite group, and vice versa.

Additionally, to evaluate the overall balance between the two samples, we estimate the propensity score (*P-Score*) and its log odds ratio (*Linearized P-Score*) using a logistic regression where the relationship indicator (*RelationshipDummy*) is regressed on both continuous and categorical covariates. By

¹⁹ The relatively low rates of bounced checks (3%) and previous loan defaults (12%), compared with the 3-month mortgage default rate (27%), do not indicate a greater propensity to default on mortgages. Notably, these differences narrow when considered against the 6-month mortgage default rate (6%), which entails more severe penalties.

calculating the overlap measures, i.e. the normalized difference, dispersion difference, and coverage frequency for the estimated propensity scores, we can assess the balance between the two groups of borrowers.

[Insert Table 2 About Here]

As demonstrated in Table 2, the distributions of all variables are nearly identical between the two samples. The statistics for the normalized difference and dispersion difference for all variables are sufficiently close to zero. Additionally, the coverage frequency at the 95% level ($\alpha = 5\%$) is also close to zero for all variables. The normalized difference, dispersion difference, and coverage frequency measures for *P-Score* in sample of the borrowers without prior relationship (with prior relationship) are 0.242, 0.097, and 0.047 (0.064), respectively. These statistics for the *Linearized P-score* are 0.241, 0.081, and 0.047 (0.064).

The normalized difference and dispersion difference for *Loan Origination Time* are both close to zero, indicating that the distribution of this variable is nearly identical across the two samples. This suggests that Maskan Bank does not differentiate between loan applications based on the existence of a prior relationship with the bank.

The balanced distribution of both individual covariates and the estimated propensity score (including the log odds ratio of the propensity score) ensures that the two sub-sample of borrowers are (hopefully) similar in unobservable factors, and our inference is not based on extrapolation due to a lack of corresponding borrowers in the other sample.

5. Results

In this section, we discuss the results of estimation of Equation (1) using LPM, probit, and logit estimation techniques as our statistical methods for exploring whether there is any linkage between prior banking relationships and borrower's incentives in payment of loan installments. The average marginal effects are presented in columns (1), (2), (3), and (4) of Table 3, respectively.

[Insert Table 3 About Here]

Our main variable of interest, the *RelationshipDummy*, exhibits a negative effect across all three models, indicating that borrowers with pre-existing relationships with the bank are less likely to default on their loans compared to those without such relationships. Specifically, the average marginal effect of prior relationships, as reported in columns (1) to (4), is -0.0403, -0.0404, -0.0408, and -0.0369, respectively, all of which are statistically significant at the 1% level. This indicates that having a prior relationship lowers default risk by about 4.0%, exclusively attributable to the borrower's incentives. The impact is

also economically meaningful given that the average default rate is 26.49%. The findings on borrowers' characteristics — gender (*Male*) and marital status (*Married*) — exhibit insignificant coefficients across all three models.

The results consistently indicate a significant positive association between the borrower's risk measures — *Loan to Value*, *Bounced Check*, *Default in Previous Loans*, and *Effective Rate* — and the probability of default across all models. Specifically, a one standard deviation increase in the *Loan to Value* ratio increases the probability of default by approximately 1.57%. The presence of at least one bounced check in borrower's record history is associated with a substantial increase in default risk, raising it by 15.5%, 13.7%, 13.9%, and 14.9% in columns (1) to (4), respectively. The average marginal effect of having a default history in previous loans (*Default in Previous Loans*) is shown to significantly increase the likelihood of default by 49.6%, 38.3%, 37.2%, and 50.3% in columns (1) to (4), respectively. The coefficient on the *Effective Rate* suggests that one percent increase in the loan effective rate leads to 0.99%, 1.11%, 1.19%, and 0.83% increase in the default risk.

Comparing the average marginal effect of the *RelationshipDummy* with other variables reveals the substantial economic significance of having prior relationships. For instance, as shown in Table 3, the approximate 4% impact of prior relationships on default risk is notably more than the impact of *Loan to Value*. Specifically, one standard deviation increase in the *Loan to Value* only raises the default risk by 1.57%, highlighting the greater economic significance of maintaining prior relationships with the bank. However, the borrower's history, particularly records of bounced checks (*Bounced Check*) and previous loan defaults (*Default in Previous Loans*), emerges as more economically significant, increasing the probability of default by an average of 14.68% and 42.65%, respectively, across all models.

In summary, the findings derived from the estimations across the three columns consistently highlight that prior relationships between the borrower and the lender significantly reduces the borrower's inclination to partake in moral hazard, thereby decreasing the likelihood of defaulting on the loan. These outcomes emphasize the protective function of pre-existing relationships with the lending institution in reducing default occurrences, underscoring the value of relationship banking for borrowers.

Our findings align with the research conducted by [Schoar \(2012\)](#), which shows that borrowers who receive heightened personalized engagement with the bank demonstrate significantly better loan repayment, characterized by decreased delinquency rates, fewer delinquency occurrences, and a delayed onset of initial default. Furthermore, [Puri et al. \(2017\)](#) exhibit a reduction in borrower's incentives to default due to the perceived value of the relationship, a finding consistent with our results. The results also align with existing literature ([Schäfer, 2019](#)), which indicates that borrowers strategically choose to default on banks that pose fewer consequences for them.

6. Robustness Checks

In this section, we conduct a battery of robustness tests. First, we use alternative estimation methods including Cox proportional hazard model as a semi-parametric survival analysis and machine learning estimation methods such as random forest, XGBoost, and neural network. Second, we address endogeneity concerns in three ways. We explore whether establishing relationship with Bank Maskan is mainly derived by geographical distance. This provides suggestive evidence that the assignment of relationship with the bank is exogenous to the individual characteristics. Next, we control for borrowers' wealth, by using a sub-sample of borrowers domiciled in Tehran, to make sure that our results are not driven by differences in wealth. Last, we address the endogeneity concern by limiting the sample of individuals to those who already have relationships with Maskan Bank. This approach helps us exclude the possibility that our results are driven by unobservable differences between borrowers with prior relationships and those without. Third, we examine the robustness of our results using alternative default measures.

6.1. Alternative Estimation Methods

6.1.1. Survival Analysis

In our baseline investigation, we employ LPM, probit, and logit regressions with the full observation period default horizon as the dependent variable. In this subsection, we use a different approach that seeks to examine how long it will take for a default to occur during the study period. We primarily employ the Cox proportional hazard model (Cox, 1972) for our analysis in this subsection to tackle and overcome the right-censorship in data.

Cox proportional hazard regression is particularly suited for analyzing time-to-event data. It is a widely adopted technique to explore the hazard rates of loan default among borrowers (e.g. Ongena and Smith, 2001; Baele et al., 2014; Agarwal et al., 2018). By estimating hazard ratios, we can assess the impact of various factors on default risk, providing valuable insights into the determinants of loan default behavior among borrowers. We follow the extant literature, and adopt the following specification:

$$h(d | x) = h_0(d) \cdot \exp \left(\beta_1 \text{RelationshipDummy}_i + \beta_2 \text{Characteristics}_i + \beta_3 \text{RiskMeasures}_i + \sum_{t=1}^M \beta_{4t} T_t \right) \quad (2)$$

Where i and t subscripts represent borrower and loan origination month, respectively.

This semi-parametric model allows for a non-parametric baseline hazard rate as well as potentially time-varying explanatory variables. In this model, the dependent variable is time-to-default (d) where default is defined as failure in repaying installments for three consecutive months. Similar to Equation (1), our main variable of interest is the $\text{RelationshipDummy}_i$. The hazard ratio associated with this variable

provides a measure of the comparative default risk between borrowers with a prior relationship and those without such a relationship. The control variables are the same as those used in Equation (1).

Full repayment of a loan or the conclusion of the sample period can result in the absence of default observation for that particular loan. In the latter case, such loans are considered “right-censored”, as whether the borrowers ultimately default on them remains unknown. In the presence of right-censoring, accurately estimating the time to default becomes problematic and failure of controlling for right censoring can result in biased and inconsistent estimates of the model parameters. To tackle this issue, right-censored observations could be incorporated in the estimation by formulating the log-likelihood function as a weighted combination of the sample density of loans experiencing defaults in the study period, and the survivor function representing the right-censored loans. This approach ensures that both defaulted and censored loans contribute to the estimation process, yielding consistent parameter estimations.

Based on the information about the loan’s initial installment date and the monthly loan performance, we construct a survival data where the loan contract date marks the initial time point and the occurrence of the first loan default represents the event of interest. Each individual (borrower) is enrolled and followed for a specified duration, denoted as the subject’s follow-up period. According to our dataset, the shortest follow-up period until the first default is observed at three months, whereas the longest duration until a borrower defaults for the first time is 23 months post-loan acquisition (see Table 4). As indicated in Table 4, a total of 109,706 borrowers had no default incidents during their follow-up period, with an average duration of 11.6 months. However, 39,524 borrowers had at least one default incident during their follow-up period, with an average duration of 6.75 months.

[Insert Table 4 About Here]

To gain a preliminary understanding of the dataset, Figure IA1 in the Online Appendix depicts the time-to-event follow-up periods for a random sample of 50 borrowers, indicating whether they default or depart the study window with a clean payment record. As we can see, all subjects remain in the dataset until the conclusion of the period unless a default event is recorded. This suggests that the sole reason for right censoring in our dataset is the conclusion of the study period. It is also important to highlight that our survival data exclusively demonstrates right censoring, with no instances of left or interval censoring, as we consistently observe loans originated after March 2019 on a monthly basis without any missing data points during the borrowers’ follow-up period. Additionally, there are no truncations in our data. It is important to note that while this figure showcases only a small sample of our data, the aforementioned characteristics hold for the entire dataset.

An intriguing aspect of the data is the duration until the first default. The data reveals that the highest number of defaults happens between 3 and 4 months after the borrower acquires the loan, afterwards, the frequency of first defaults decreases as time passes. Figure 1 presents a histogram for the two groups

of borrowers, one who have a default during the study time and one who leave the study period without any defaults. It is evident that there are relatively few borrowers who manage to avoid default as time elapses. The majority of these borrowers remain in the dataset for 11 to 13 months before they exit the study window.

[Insert Figure 1 About Here]

Table 5 presents the results of our analysis. In column (1), we focus solely on the variable of interest, the *RelationshipDummy*, which is included as the explanatory variable. In column (2), we expand the model by introducing control variables that capture borrowers' characteristics, such as gender (*Male*) and marital status (*Married*), as well as fixed effects for job and education. Finally, in column (3), we further enhance the analysis by including additional variables that control for borrowers' risk measures, specifically *Loan to Value*, *Bounced Check*, *Default in Previous Loans*, and *Effective Rate*.

The hazard ratios for *RelationshipDummy* are 0.818, 0.844, and 0.807, as reported in columns (1), (2), and (3) of Table 5, respectively, all of which are statistically significant at the 1% level. These ratios indicate that borrowers who have a savings or current account before the mortgage origination date have hazard rates that are 18.2% ($= 1 - 0.818$), 15.6% ($= 1 - 0.844$), and 19.3% ($= 1 - 0.807$) lower than those of borrowers without such a relationship. These findings re-affirm the robustness of our previous results from LPM, probit, and logit regressions, highlighting that maintaining a prior relationship with the bank serves as a protective factor against default risk.

[Insert Table 5 About Here]

The findings on the control variables consistently reveal a positive association between default and factors such as *Loan to Value*, *Bounced Check*, *Default in Previous Loans*, and *Effective Rate* across all models. These results are qualitatively consistent with those observed in the main model, reaffirming the significance of these risk measures. On the other hand, control variables like *Male* and *Married* exhibit insignificant coefficients in column (3), aligning with our main findings.

While the results of Cox model present promising findings, it is imperative to verify whether the hazard ratios maintain proportionality over time. This assessment is critical to bolster our confidence in the validity and reliability of our conclusions. Thus, we investigate whether the proportional hazards assumption holds (Hosmer et al., 2008). This assumption posits that the hazard ratios for the covariates remain constant over time, while the hazard rates are allowed to vary over time. Violation of this assumption in Cox regression analysis can result in biased estimates (Kuitunen et al., 2021). Our tests confirm that the proportional hazards assumption holds.²⁰

²⁰ We use several types of tests such as: comparison of observed and predicted survival curves (Cleves et al., 2008), checking parallelness of *log-log* survival curves, checking parallelness of adjusted *log-log* survival curves,

Overall, the consistent findings of the Cox model provide robust evidence of the relationship between borrower’s prior relationships and default behavior, enhancing the validity and reliability of our previous results.

6.1.2. Machine Learning Methods

In this subsection, we extend our analysis by applying machine learning techniques to estimate the primary model specified in Equation (1). We utilize machine learning models — specifically, random forest (Breiman, 2001; Khandani et al., 2010), XGBoost (Chen and Guestrin, 2016), and neural networks (Ciampi et al., 2013) — which have demonstrated success and popularity in predictive tasks. Following the methodology of Fuster et al. (2022), we employ these models to assess their predictive power and robustness in estimating default risk compared to the logit model.²¹

We begin by estimating all four models — logit, random forest, XGBoost, and neural network — using a portion of our full dataset, designated as the training set. To evaluate the models’ performance, we use a separate test set. Specifically, 70% of the dataset is randomly selected for training, while the remaining 30% is reserved for testing. This random sampling is performed across all loans, ensuring that the division between training and test sets is independent of all characteristics.

Within the training set, we further split the data into two subsets. The first subset, representing 70% of the training data, is used to estimate all models and train the machine learning models. The second subset, which comprises the remaining 30% of the training data and is referred to as the calibration sample, is utilized to apply isotonic regression. This step refines the predicted class labels from the machine learning models into calibrated probabilities. By employing this approach, we ensure that both the traditional and machine learning models are based on an equivalent amount of data for estimating default probabilities.²²

Unlike the logit model, which allows for straightforward interpretation of marginal effects due to its parametric nature, machine learning models such as random forest, XGBoost, and neural networks are inherently non-parametric. This non-parametric nature introduces complexity in understanding direct marginal effects, as these models capture intricate, non-linear interactions between features. Consequently, while interpreting direct marginal effects is challenging, we analyze the change in the probability of default by evaluating the impact of switching dummy variables from zero to one and

and graphical test of scaled Schoenfeld residual (Schoenfeld, 1982; Grambsch and Therneau, 1994). The results of these tests are not presented in this paper but can be provided upon request from the authors.

²¹ For a comprehensive discussion on these models, please refer to the studies by Khandani et al. (2010), Breiman (2001), Fuster et al. (2022), and Ciampi et al. (2013).

²² Evaluation of the model’s performance is not presented in the paper but can be provided upon request from the authors.

assessing the effect of a one-unit change in continuous variables from their mean values. This approach enables us to compare the results of machine learning models with those from our main logit regression.

Table 6 presents the impact of various features on the probability of default. Notably, the marginal effects of *RelationshipDummy* are -3.23%, -3.95%, and -3.80% for random forest, XGBoost, and neural network models, respectively. These effects are consistent with our primary findings, which underscores the robustness and reliability of our results. This consistency generally extends to other features as well, with the exception of *Loan to Value*. While the neural network model's results align with our main logit model, both random forest and XGBoost models exhibit a negative effect for *Loan to Value*, contrasting with the positive effect observed in logit model.

[Insert Table 6 About Here]

6.2. Endogeneity Concerns

6.2.1. Random Assignment of Relationship with Maskan Bank

To ensure that the borrowers with prior relationships with Maskan Bank are not systematically different from those without relationships, we explore whether the assignment of the relationships can be considered almost random. Specifically, we test whether borrowers' proximity to Maskan Bank branches influences their likelihood of having prior relationships. If the distance to the bank is a significant factor in determining a borrower's relationship, this would support the notion that the relationship assignment is exogenous to the borrower's characteristics. In this analysis, we examine the geographical distance between a borrower's residence and the nearest Maskan Bank branch and its impact on the likelihood of having a pre-existing relationship. The idea is that borrowers living closer to Maskan Bank branches are more likely to engage with the bank simply due to convenience, rather than any inherent characteristics or selection biases.²³ This approach allows us to demonstrate that the prior relationships are likely the result of geographical factors, not selective treatment, thus supporting the almost random assignment of relationships.

²³ We acknowledge that neither borrower nor branch locations are randomly assigned. Accordingly, a potential concern that may arise is that Maskan Bank may have strategically located branches in wealthier districts. However, Tehran's long-run urban expansion mitigates this concern. Over recent decades, the city has grown outward from its historical center toward the north, south, east, and west. Neighborhoods that were relatively affluent in earlier periods were concentrated in central areas, whereas more recent urban and highway developments have allowed many higher-income households to relocate and move predominantly toward northern districts. Maskan Bank's branches, nonetheless, expanded gradually alongside this growth to ensure population coverage and administrative access, rather than to target contemporaneously affluent neighborhoods, given the relatively high transaction costs of changing branch network. As a result, spatial proximity between borrowers and branches primarily reflects historical urban development and subsequent residential sorting, rather than endogenous branch placement based on borrower credit quality.

We follow two approaches to analyze the link between having banking relationships and geographical distance between borrowers and Maskan Bank branches: (i) we count the number of Maskan Bank branches within circles of varying radii centered on the borrowers' neighborhoods, identified by their 5-digit ZIP codes, and (ii) we calculate the average distance to the k nearest branches. We use Google Maps to identify the geographic coordinates of each bank branch, as listed on the banks' website. In Tehran, there are 32 operating banks, each with branches distributed throughout the city. The total number of bank branches in Tehran is 1,686, with 84 branches of Maskan Bank. Figure 2 illustrates the distribution of Maskan Bank branches in Tehran.

[Insert Figure 2 About Here]

Table 7 presents the distribution of bank branches (Maskan Bank and other banks) within varying radii (500, 1000, 1500, and 2000 meters) centered on borrowers' neighborhoods center. On average, borrowers have 0.20, 0.71, 1.41, and 2.38 Maskan Bank branches within 500, 1000, 1500, and 2000 meters, respectively. In contrast, there are significantly more branches of other banks, with an average of 2.97 branches within 500 meters, increasing to 41.56 branches within 2000 meters. This disparity highlights the relatively lower density of Maskan Bank branches compared to other banks in the borrowers' vicinity.

[Insert Table 7 About Here]

We first examine the effect of having additional Maskan Bank branches in the vicinity of a borrower's residence on the likelihood of having prior relationships with Maskan Bank. To do this, we define a variable that represents the number of branches within a circle centered on the borrower's 5-digit ZIP code center, with radii of 500, 1000, 1500, and 2000 meters. We then regress the *RelationshipDummy* on the number of nearby Maskan branches (NMB_i) and other borrower characteristics. These characteristics include gender, marital status, record of default in previous loans and bounced checks, job, and education, using the following logit regression model:²⁴

$$P(\text{RelationshipDummy}_i) = \beta_0 + \beta_1 NMB_i + \beta_2 \text{Characteristics}_i + \sum_{t=1}^M \beta_{3t} T_t + \alpha_d + \epsilon_i \quad (3)$$

Where i and t subscripts represent borrower and loan origination month, respectively. To address the concern that branch expansion may be systematically concentrated in wealthier districts, we also include district fixed effects (α_d), which control for all time-invariant characteristics at the district level that could otherwise confound the relationship between branch proximity and borrower outcomes.²⁵

²⁴ For the sake of simplicity, we only report the results of logit regression. We confirm that we find similar results when we use LPM and probit regressions in the present subsection.

²⁵ The concern regarding endogenous sorting of bank branches—and, consequently, endogenous relationship formation—could, in principle, be addressed by exploiting variation in branch openings and closures over time. If such data were available, we could directly test whether branch openings are systematically related to district-

The results presented in Table 8 reveal a statistically significant positive relationship between the number of nearby Maskan Bank branches and the likelihood of a borrower having a prior relationship with the bank. In all four models (using the four different radius circles), the coefficient on the number of Maskan Bank branches is positive and statistically significant. Specifically, for the 500m circle, the presence of an additional branch increases the likelihood of a prior relationship by 1.5%. As the radius expands to 1000m, 1500m, and 2000m, the effect slightly decreases, with coefficients of 0.0632, 0.0759, and 0.0781, respectively. This suggests that the proximity of Maskan Bank branches is a crucial factor in the formation of prior relationships with borrowers, emphasizing that geographic distance plays a significant role in relationship assignment.

[Insert Table 8 About Here]

Instead of using the number of nearby Maskan Bank branches, we also examine the effect by employing another variable (MBD_i), which determines the average distance to the k nearest branches, where k belongs to the set $\{1, 2, 3, \dots, 10\}$. We define MBD_i as:

$$MBD_i \equiv \frac{\sum_{j=1}^k (\text{Distance to the } j_{th} \text{ Nearest Maskan Branch})_i}{k} \quad (4)$$

Table IA1 in the Online Appendix provides summary statistics for the distance between borrowers' neighborhoods and the nearest bank branches, including Maskan Bank and other banks. On average, the distance to the nearest Maskan Bank branch is 1,287.32 meters, while the average distance to the 10 nearest branches increases to 3,519.98 meters. The results indicate that Maskan Bank branches tend to be located farther from borrowers' neighborhoods compared to other banks, highlighting the relatively lower accessibility of Maskan Bank services.

We run Equation (3) again except we substitute NMB_i with MBD_i variable. This approach allows us to analyze how varying distances to the closest Maskan Bank branches impact the likelihood of borrowers establishing prior relationships with the bank, providing a more nuanced understanding of the influence of geographic proximity on relationship formation.

The results displayed in Table 9 reveal a statistically significant negative relationship between the distance to the nearest Maskan Bank branch (MBD) and the likelihood of a borrower having a prior relationship with the bank. In all columns, the coefficient for the distance to the nearest branches is

level wealth, local house prices, or proxies for ex ante default risk, and further examine whether changes in branch proximity affect relationship formation and subsequent repayment behavior. However, we are unable to implement this approach because data on branch opening and closure dates are unavailable. Instead, we mitigate this concern in several complementary ways. First, in this Subsection, we include district fixed effects, which absorb all time-invariant district-level characteristics, including local wealth and baseline risk. Second, we control for district-level wealth as a proxy for borrower quality (Subsection 6.2.2). Third, we show that the estimated relationship effect persists across neighborhoods within different home-price brackets (Subsection 7.3). Taken together, these approaches could alleviate concerns that Bank Maskan systematically locates branches in neighborhoods with borrowers of higher quality in terms of wealth and ex ante default risk.

negative and significant at the 1% level. The coefficients range from -0.00174 to -0.00215 as we consider the average distance to the nearest branch up to the ten nearest branches. This indicates that as the distance to the Maskan Bank branch increases, the likelihood of establishing a prior relationship decreases.

[Insert Table 9 About Here]

In addition to our logit regression analysis, we employ the machine learning method of random forest to assess the feature importance of each variable, especially the borrowers' distance to the Maskan branches, in predicting whether a borrower has a prior relationship with Maskan Bank. By analyzing the importance scores generated by the model, we identify which factors, including the distance to Maskan branches, gender, marital status, and prior loan defaults, significantly influence the likelihood of borrowers establishing a relationship with the bank.

The results from the random forest analysis, as shown in Figure IA2 in the Online Appendix, reveal the feature importance scores for predicting whether a borrower has a prior relationship with Maskan Bank. The most significant predictor is the *Distance to Maskan Branch (MBD)*, with an importance score of 0.835, indicating that the distance to branch plays a crucial role in determining borrower relationships. Following *MBD*, *Job* has a much lower importance score of 0.081, suggesting that it also contributes to the prediction but to a significantly lesser extent. Other features, such as *Education* (0.057) and *Gender* (0.009), demonstrate even lower levels of importance. The variables *Marital Status* (0.006), *Default in Previous Loans* (0.005), and *Bounced Check* (0.004) have minimal impact on the model's predictions. These findings underscore the dominance of proximity to Maskan Bank branches in influencing borrowers' relationships with the bank.

The presence of other banks branches in the vicinity of borrowers' neighborhoods is also an important issue which can affect the borrowers' decision for having relationship with Maskan Bank. To assess this case, we introduce a ratio (DR_i) defined as follows:

$$DR_i \equiv \frac{\text{Distance to the Nearest Maskan Branch}_i}{\text{Average Distance to the } k \text{ Nearest Other Banks Branches}_i} \quad (5)$$

Where $k \in \{1, 2, 3, \dots, 10\}$. We hypothesize that an increase in the distance to Maskan Bank branches or a decrease in the distance to other banks' branches will lead to a lower probability of borrowers establishing a relationship with Maskan Bank. Consequently, we run Equation (3) again, substituting NMB_i with the DR_i ratio.

[Insert Table 10 About Here]

The results presented in Table 10 illustrate a statistically significant negative relationship between the distance ratio (DR) and the likelihood of borrowers having a prior relationship with Maskan Bank. The coefficient on DR consistently indicates a significant negative impact, with estimates ranging from

-0.0023 for $k = 1$ to -0.0101 for $k = 10$. This suggests that as either the distance to the nearest Maskan branch increases or the distance to the other banks' branches decreases, the probability of establishing a prior relationship with Maskan Bank declines.

We also employ the random forest to assess the feature importance of variables, particularly focusing on the *DR*, in predicting whether a borrower has a prior relationship with Maskan Bank. The results as illustrated in Figure IA3 in the Online Appendix shows that *DR* is the most important predictor, with importance scores ranging from 0.8306 to 0.8415 across different models with different values of k . In comparison, other variables such as *Job* (0.0797 to 0.0879), education (0.0528 to 0.0586), and *Gender* (0.0082 to 0.0091) show significantly lower importance scores. *Marital Status* and other historical variables, such as *Bounced Checks* and *Defaults in Previous Loans*, exhibit even smaller importance values.

It is important to distinguish between the incentives to form a relationship with a bank and the incentives to maintain it. Relationships accumulate over time, creating strong incentives for clients to preserve them. However, a borrower's *initial* relationship with a specific bank is largely determined by the geographic proximity of the borrower's residence to bank branches. This suggests that the choice of which bank a borrower first engages with can be treated as approximately exogenous to both observable and unobservable borrower characteristics. Importantly, showing that proximity drives initial assignment does not diminish the value of the relationship itself; once formed, borrowers have strong incentives to maintain these relationships, consistent with the central argument of this paper that relationships create a borrower-driven channel affecting repayment behavior.

6.2.2. Controlling for Borrower's Wealth

To ensure the robustness of our findings, it is essential to account for the potential influence of borrower's wealth on the link between prior relationships and default behavior. Specifically, there may be a correlation between having prior relationships with Maskan Bank and borrower's wealth, which could affect the coefficient on the *RelationshipDummy*. In this subsection, we utilize the average home price per square meter in the borrowers' neighborhoods at the time of loan origination as a proxy for wealth, since we do not have direct data on borrowers' wealth and income. The assumption is that individuals living in more expensive areas tend to be wealthier.

We obtain monthly housing price data (per square meter) in Tehran from the Ministry of Roads and Urban Development. In our sample, average prices range from \$250 to \$1,450 per square meter (see Table IA2 in the Online Appendix for details). The housing price data are reported at the 6-digit ZIP code, while our loan dataset identifies borrowers' locations using 5-digit ZIP codes. To align the two sources, we aggregate housing prices to the 5-digit ZIP level. Because these data are available only for

Tehran, the analysis in this subsection is limited to that city. Moreover, as information on the size of borrowers' homes is unavailable, we cannot compute total home values.

Another potential concern is that borrowers with prior bank relationships — who may be concentrated in certain neighborhoods — may be more risk-averse than those without such relationships. To address this, we also include district fixed effects in the specification.

The results for the *RelationshipDummy* variable, presented in Table 11, are consistent with our main findings. The negative coefficient across LPM, probit, and logit models remains statistically significant at the 1% level, indicating that borrowers with prior relationships are less likely to default. The magnitude of the effect is slightly increased, with marginal effects of approximately -4.3% across all models, which is only marginally different from the results in the main analysis. This consistency reinforces the robustness of the borrower's incentive channel, even when accounting for borrower's wealth.

[Insert Table 11 About Here]

6.2.3. *The Sample of Borrowers with Prior Relationships*

A potential concern is that the existence of a prior relationship with Maskan Bank may be endogenous. To address this issue, we restrict the analysis to borrowers who had already established a relationship with the bank before applying for a mortgage under the *Second Program*.

One might argue that if maintaining a relationship with the bank provides borrowers with incentives to avoid default, then borrowers with stronger relationships should exhibit lower default rates. Since we lack data on the intensity of relationships, we measure relationship strength by its duration. Accordingly, we replace our main variable of interest, *RelationshipDummy*, with *RelationshipDuration*, defined as the number of years from the opening of a borrower's first account with the bank to the origination of the loan. This specification allows us to estimate the marginal effect of an additional year of relationship on default behavior, conditional on having an existing relationship. The results, reported in Table IA3 in the Online Appendix, show that the coefficient on *RelationshipDuration* is statistically significant across all specifications.

This underscores the importance of the depth of the borrower-bank relationship in influencing the borrower's inclination to avoid default. The result is also consistent with [Fiordelisi et al. \(2014\)](#), which highlights the benefits of longer bank-borrower relationships in reducing the probability of default.

6.3. Alternative Default and Relationship Measures

6.3.1. Alternative Default Horizons

Our primary analysis uses a binary dependent variable to indicate whether a borrower defaults within the full observation period following loan origination, with default defined as the failure to repay installments for three consecutive months. To explore the robustness of our findings, we consider alternative default horizons. Specifically, we re-define the dependent variable to capture defaults occurring within 6-month and 12-month periods post-loan origination, again defining default as a failure to make payments for three consecutive months. We re-estimate Equation (1) with the same explanatory variables, but using the alternative definitions of dependent variable to examine the impact of different time frames on our results.

The results presented in Table IA4 in the Online Appendix indicates that the average marginal effects of having prior relationships are significantly negative across both horizons, suggesting that borrowers with prior relationships consistently have less incentives to default. These findings align closely with our outcome when considering the full observation period for default, reinforcing the robustness of prior relationships as a protective factor against default. The coefficients on the control variables are also qualitatively similar to the main results, with minimal changes.

6.3.2. Weak-Default vs Hard-Default

In our initial analysis, default is defined as being delinquent for 90 days in repayment, denoted as “weak-default”. This criterion entails marking the failure event when borrowers have surpassed a 90-day period without making any repayments on their loans. An alternative metric for default, termed “hard-default”, stipulates default as occurring after 180 consecutive days without any repayments (Puri et al., 2017). To ensure the reliability of our results, we adjust the dependent variable to the hard-default definition and re-estimate Equation (1). This method enables us to evaluate the consistency of our findings with a different default definition and validate the robustness of our primary results. Furthermore, it allows for a deeper understanding of borrowers’ behaviors concerning the two distinct types of defaults: weak-default and hard-default.

The analysis, with results reported in Table IA5 in the Online Appendix, reveals that borrowers with a pre-existing relationship demonstrate a significant disincentive toward hard-default. Specifically, the average marginal effect of having prior relationships (*RelationshipDummy*) suggests that the risk of default is approximately 1.53% to 1.59% lower for borrowers with relationships compared to those without. These results are qualitatively similar to our main findings using weak default as the

dependent variable. However, they suggest that prior relationships have a stronger economic effect in preventing weak defaults than hard defaults.²⁶

A plausible explanation is that once a hard default occurs, Maskan Bank initiates legal proceedings to seize the collateral. As delinquency escalates toward hard default, the incentive to preserve the lending relationship in order to secure future financial benefits becomes less relevant. At that stage, the primary concern shifts to minimizing the loss of pledged collateral. By contrast, borrowers are more likely to avoid weak default in order to preserve relationship capital and maintain access to future financing.

Our findings align with Schäfer (2019) in highlighting that the effects of relationship lending depend on the stage and severity of delinquency. While relationship lending may permit temporary distress (leading to higher short-term delinquency), it does not translate into more persistent or severe delinquency. Similarly, we find that relationship capital plays a stronger role in preventing mild forms of default, whereas its importance diminishes once delinquency escalates and legal enforcement becomes the dominant mechanism.

7. Further Analysis

7.1. COVID-19 Lockdown and Relationship Banking

The COVID-19 pandemic and subsequent lockdowns brought about unprecedented challenges for households worldwide, causing an exogenous income shock (Bruce et al., 2022). Such an unforeseen income shock has significant implications for borrowers' ability to meet their loan obligations. Previous studies indicate a surge in loan defaults in the immediate aftermath of the COVID-19 lockdown (Nigmonov and Shams, 2021). Our sample presents a similar trend. In Iran, the lockdown began in March 2020. Figure 3 shows the trend in the number of defaults each month, spanning 11 months prior to the lockdown period until 11 months thereafter.

[Insert Figure 3 About Here]

As we can see, following the onset of the lockdown in March 2020, there is a notable increase in defaults. This sharp rise reflects the financial strain experienced by borrowers due to pandemic-related disruptions to household income and economic activity. However, as time progresses beyond the initial shock period, there is a gradual decline in defaults, suggesting a diminishing trend in default rates. This

²⁶ In addition to the lower rates of default, we find that borrowers with a prior relationship exhibit a delinquency (monthly delinquency) rate approximately three percentage points lower than those without such a relationship.

decrease may signify a partial recovery or adaptation of borrowers to the new economic realities imposed by the pandemic.

Within the realm of banking, the role of pre-existing relationships between borrowers and financial institutions during such crises is unclear priori. Indeed, there are two competing hypotheses:

Positive Effect: Borrowers may perceive their pre-existing relationships with the bank as valuable assets during the pandemic. They may strive to preserve these relationships, anticipating potential future benefits in accessing financial support if the crisis worsens. This perspective suggests that borrowers with established relationships may be more inclined to honor their loan commitments despite the financial difficulties induced by the pandemic.

No Effect: Conversely, pre-existing relationships might not significantly influence borrowers' default decisions during the pandemic lockdown. Borrowers facing financial distress may view the crisis as a justification for defaulting, regardless of their relationship status with the bank. This perspective implies that the impact of relationship banking on loan default behavior remains negligible in the face of severe economic disruptions.

To empirically examine the influence of pre-existing relationships on loan default behavior during the COVID-19 lockdown, we conduct an event-study to incorporate the pandemic's temporal dynamics.²⁷ The regression model takes the following form with the same explanatory variables in Equation (1), except that we include the interaction term between *RelationshipDummy_i* and month dummies (*T_t*):

$$P(Default)_{it} = \beta_0 + \sum_{t=0}^{t=23} \beta_{1t} T_t + \beta_2 RelationshipDummy_i + \sum_{t=0}^{t=23} \beta_{3t} RelationshipDummy_i \times T_t + \sum_{t=0}^{t=23} \beta_{4t} Characteristics_{it} + \sum_{t=0}^{t=23} \beta_{5t} RiskMeasures_{it} + \epsilon_{it} \quad (6)$$

Where *i* and *t* subscripts represent borrower and time, respectively.

In this regression, *T_t* represents a vector of dummy variables for month *t*. It begins from 2019-04 and ends in 2021-02. The benchmark time is the beginning of the quarantine in March 2020. By incorporating the interaction term between *RelationshipDummy_i* and *T_t*, we aim to elucidate the nuanced dynamics of relationship banking amidst the COVID-19 lockdown, shedding light on its implications for borrower default behavior.

First, we examine whether a parallel trend exists in the default behavior of borrowers with prior relationships and those without, before the inception of the lockdown in March 2020. Figure 4 indicates

²⁷ We use a panel linear probability model in this subsection. Also, we estimate the model using probit and logit regressions; however, for the sake of simplicity we only report the results of LPM regression. We confirm that we find similar results when we use probit and logit regressions.

that all coefficients on the interaction term for the period after June 2019 and prior to March 2020 are statistically insignificant, confirming the parallel trend assumption.

[Insert Figure 4 About Here]

The graph also reveals an interesting trend amid the overall increase in loan defaults during the three months following the lockdown: borrowers with prior relationships with the bank exhibit significantly lower default rates compared to those without such relationships. This finding suggests that maintaining pre-existing relationships may mitigate default risk during periods of economic uncertainty and financial distress.

7.2. *Natural Disasters and Relationship Banking*

Studying borrowers' behavior and their incentives to maintain banking relationships is particularly valuable in the context of catastrophic events, especially as climate change has increased the frequency and intensity of natural disasters (Usman, 2016; Kim and Lee, 2023; Bohnhoff et al., 2024). Natural disasters pose significant threats to communities by causing widespread damage to infrastructure and by adversely affecting quality of life. These challenges require proactive strategies to adapt to such risks and ensure future well-being (Yu et al., 2019). One key adaptation mechanism for households is the use of financial services and borrowing (Zeng et al., 2021).

Maintaining a strong relationship with banks hence becomes particularly valuable during periods of distress, as it facilitates easier access to credit (Abedifar et al., 2024). Banks can provide timely financial assistance, support long-term adaptation efforts, and help individuals overcome challenges posed by natural disasters. Consequently, borrowers who have suffered from catastrophic events are more likely to preserve their banking relationships for potential future use. Guided by these considerations, this subsection examines how the occurrence of an earthquake influences the value individuals place on their banking relationships.

The catastrophic event analyzed in this study is the earthquake that struck West Azerbaijan and East Azerbaijan provinces on February 23, 2020. According to the Iranian Seismological Center (IRSC), this earthquake registered a magnitude of 5.7 on the Richter scale, with its epicenter near the city of Khoy in West Azerbaijan. The quake occurred at a depth of 6.4 kilometers and was strongly felt in nearby cities, including Tabriz, Osku, and Urmia. The Iranian Red Crescent Society reported that the earthquake impacted 43 villages, resulting in 10% to 100% destruction of about 12,000 residential buildings in the Khoy region and severe injuries to 75 people.

To investigate the influence of prior banking relationships on loan default behavior in the aftermath of the earthquake, this study employs a triple difference (difference-in-difference-in-differences) methodology. Borrowers residing in the earthquake-affected counties (Khoy, Urmia, Osku, and Tabriz)

directly impacted by the quake, are classified as the treatment group, comprising 4,659 borrowers. To construct a counterfactual group, 5,856 borrowers from counties in neighboring provinces (Ardabil, Kermanshah, and Kurdistan) are selected as the control group.

[Insert Table 12 About Here]

Table 12 shows that the distributions of all variables are highly similar between the treatment and control groups, confirming the robustness of our selection process. The normalized difference and dispersion difference for all variables are close to zero, as expected. Additionally, the coverage frequency at the 95% confidence level ($\alpha = 5\%$) is also close to zero across variables. For the sample of borrowers in the treated group (and the control group), the normalized difference, dispersion difference, and coverage frequency measures for the *P-Score* are 0.089, 0.022, and 0.05 (0.05), respectively. For the *Linearized P-score*, these statistics are 0.090, 0.021, and 0.05 (0.05), respectively.

We adopt the following panel specification, focusing on the period from September 2019 (five months before the shock) to July 2020 (five months after the shock):

$$\begin{aligned}
P(Default)_{it} = & \beta_0 + \sum_{t=0}^{t=11} \beta_{1t} T_t + \beta_2 Quake_i + \beta_3 RelationshipDummy_i + \sum_{t=0}^{t=11} \beta_{4t} T_t \times Quake_i + \\
& \sum_{t=0}^{t=11} \beta_{5t} T_t \times RelationshipDummy_i + \beta_6 Quake_i \times RelationshipDummy_i + \\
& \sum_{t=0}^{t=11} \beta_{7t} T_t \times Quake_i \times RelationshipDummy_i + \\
& \sum_{t=0}^{t=11} \beta_{8t} Characteristics_{it} + \sum_{t=0}^{t=11} \beta_{9t} RiskMeasures_{it} + \epsilon_{it}
\end{aligned} \tag{7}$$

Where i and t subscripts represent borrower and time, respectively.

In this regression, T_t represents a vector of dummy variables for each month t , spanning the period from September 2019 (2019-09) to July 2020 (2020-07). The benchmark month is February 2020 (2020-02), the month of earthquake.²⁸ $Quake_i$ denotes the treatment dummy which equals one if the borrower resides in the treated counties and zero otherwise. The coefficient on the triple interaction term among T_t , $Quake_i$, and $RelationshipDummy_i$ captures the effect of having a prior relationship with the bank on the default rate in the earthquake-affected (treated) counties, relative to unaffected (control) counties, during the specified timeframe.

²⁸ The COVID-19 lockdown began in March 2020 (2020-03); however, this should not be a concern for our analysis as it affected both treatment and control groups equally. Furthermore, this is validated by the pre-earthquake parallel trend.

[Insert Table 13 About Here]

Table 13 presents the results. To validate one of the key assumptions of the difference-in-differences model, we first assess whether the pre-treatment parallel trend assumption holds. The insignificant triple interaction terms for the five months preceding the earthquake indicate that the treated and control groups exhibit parallel trends before the event, thereby confirming that this assumption is satisfied.

The estimated coefficient on the triple interaction term for the first month after the earthquake, March 2020, is significantly negative at the 5% and 10% levels across all three specifications. The magnitude of the effect is economically meaningful: having a prior relationship with the bank reduces the default rate by 1.31, 1.56, and 1.29 percentage points in columns (1), (2), and (3), respectively. These reductions are substantial, given that the average monthly default rate before the earthquake in both treatment and control groups is 2.97%.²⁹

The lack of significance for the triple difference terms in subsequent months may be attributed to the substantial government aid provided after the earthquake and the absence of further seismic events, both of which likely reduce borrowers' reliance on banking relationships.³⁰

7.3. *The Role of Wealth in Maintaining Banking Relationship*

We demonstrate that borrowers tend to pay mortgage installments to preserve the relationship that they have built with the bank. The value of the relationship may depend on its reusability. Hence, borrowers who are more likely to use their banking relationship in the future may have a greater incentive to avoid default. In this context, we investigate the role of borrowers' wealth in loan performance, comparing those with a prior relationship to those without one. On the one hand, wealthier borrowers may place less value on the banking relationship because they possess strong "hard" information that can support their future credit applications (Liberti and Petersen, 2019). On the other hands, such borrowers may attach more value to the relationship because they are more likely to use it in the future, given they may have a greater and wider range of opportunities.

²⁹ The average monthly default rate is calculated by dividing the number of defaults by the number of borrowers each month during the period preceding the earthquake for both the treatment and control groups and then averaging these monthly rates. This figure differs from the overall average default rate in the sample, which is 26.49% (Table 1). The latter is computed by dividing the total number of borrowers who recorded a default by the total number of borrowers across the entire dataset.

³⁰ The level of statistical noise may vary across the estimation period. Pre-earthquake months reflect relatively stable conditions, resulting in low volatility in month-specific coefficients. In contrast, the immediate post-earthquake period involves substantial disruption, which can increase noise in repayment behavior and widen confidence intervals. Additionally, the government provided substantial financial aid shortly after the earthquake, which likely reduced borrowers' reliance on relationship-based credit and further attenuated the estimated effects, contributing to greater dispersion in post-event coefficient estimates.

Our aim is to examine the value of relationship among borrowers with different levels of wealth. We estimate the following regression model for our analysis:

$$P(Defauli)_i = \beta_0 + \beta_1 RelationshipDummy_i + \beta_2 HomePrice_i + \beta_3 (RelationshipDummy_i \times HomePrice_i) + \beta_4 Characteristics_i + \beta_5 RiskMeasures_i + \sum_{t=1}^M \beta_{6t} T_t + \sum_{d=1}^D \beta_{7d} D_d + \epsilon_i \quad (8)$$

Where i and t subscripts represent borrower and loan origination month, respectively.

We define *Default* as a function of a dummy variable indicating prior relationships with the bank (*RelationshipDummy_i*), home price (*HomePrice*), the interaction term between these two variables (*RelationshipDummy_i × HomePrice_i*), and the control variables used in our main analysis (represented by *Characteristics_i* and *RiskMeasures_i*, time and regional fixed effects). The interaction term is our variable of interest. The analysis is limited to mortgage loan data for Tehran because home price is only available for this city.

Figure 5 display the results. The marginal effects of prior relationships at different home prices are statistically significant at the 1% level (please refer to Table IA6 in the Online Appendix for more details).³¹ The results indicates that borrowers with higher wealth — those in more expensive neighborhoods — have stronger incentives to avoid default. The marginal effects range from -2.96% to -7.89%. This implies that the default probability of the borrowers — with a prior relationship with the bank — living in a neighborhood with an average home price per square meter of \$250 (\$1,450) is 2.96% (7.89%) less than the default probability of the borrowers without such a relationship living in the same neighborhood.

[Insert Figure 5 About Here]

Overall, the analysis reveals a geographic disparity in the borrowers' incentives channel. Borrowers residing in Tehran's wealthier districts (the upper regions of the city as shown in Figure 6) demonstrate a stronger response to the potential benefits of maintaining a relationship with the bank. This reinforces the notion that the incentive to avoid default is heterogeneous across wealth levels.

[Insert Figure 6 About Here]

7.4. The Role of Gender and Marital Status Maintaining Banking Relationship

The existing literature documents that women face discrimination in access to credit (Bellucci et al., 2010; Alesina et al., 2013; Ongena and Popov, 2016), despite evidence showing that female borrowers do not exhibit higher delinquency rates (Beck et al., 2018). Moreover, Abedifar et al. (2024) show that

³¹ Additionally, the difference between the marginal effects at different home prices are statistically significant at the 5% level.

banking relationships generate larger benefits for female borrowers in terms of access to new credit. Taken together, these findings suggest that the value of maintaining a banking relationship may be particularly high for women. This raises an important question: do female borrowers have stronger incentives to avoid default in order to preserve these relationship benefits?

This question is worth exploring because, on the one hand, banking relationships may be more valuable for female borrowers than male borrowers. On the other hand, it can be argued that nurturing banking relationships might be even more important for male borrowers, as they are more likely to engage in business activities as part of their long-term career and financial goals (Hira and Mugenda, 2000). Consequently, male borrowers might be more inclined to maintain their banking relationships. Indeed, major financial milestones such as home ownership, business investments, or large-scale purchases typically require solid credit histories. Men aiming for these goals are more likely to preserve their creditworthiness to secure favorable loan terms and interest rates. Furthermore, maintaining a robust credit profile enhances their future financial leverage, granting access to better financial opportunities. This approach helps them capitalize on future prospects and minimize the risks associated with poor credit, thereby strengthening their commitment to avoid default.

Marital status may further shape these incentives. Prior research suggests that married individuals exhibit higher levels of commitment and responsibility in professional and organizational settings (Hudis, 1976; Bielby, 1992; Kacmar et al., 1999; Çemberci et al., 2022). If such commitment extends to financial behavior, married borrowers may place greater value on maintaining banking relationships and thus display stronger incentives to avoid default. Given these competing mechanisms, the net effect of gender and marital status on the relationship-incentives channel is theoretically ambiguous and warrants empirical investigation.

To examine this issue, we estimate Equation (9) using linear probability, probit, and logit models. In addition to the baseline controls from Equation (1), the specification includes interaction terms between *RelationshipDummy* and borrower group dummies — *Single Male*, *Married Male*, and *Single Female* — using *Married Female* borrowers as the reference category. To account for potential differences in wealth and income across groups, we include neighborhood-level home prices as a proxy for borrower wealth. In addition, regional fixed effects are incorporated into the model. The specification is as follows:

$$\begin{aligned}
 P(\text{Default})_i = & \beta_0 + \beta_1 \text{RelationshipDummy}_i + \beta_2 (\text{RelationshipDummy}_i \times \text{Single Male})_i + \\
 & \beta_3 (\text{RelationshipDummy}_i \times \text{Married Male})_i + \beta_4 (\text{RelationshipDummy}_i \times \text{Single Female})_i + \\
 & \beta_5 \text{Single Male}_i + \beta_6 \text{Married Male}_i + \beta_7 \text{Single Female}_i +
 \end{aligned} \tag{9}$$

$$\beta_8 \text{ Characteristics}_i + \beta_9 \text{ RiskMeasures}_i + \sum_{t=1}^M \beta_{10t} T_t + \sum_{d=1}^D \beta_{11d} D_d + \epsilon_i$$

Where i and t subscripts represent borrower and loan origination month, respectively.

[Insert Table 14 About Here]

The estimation results using LPM, probit, and logit regressions are presented in columns (1), (2) and (3) of Table 14, respectively. Prior banking relationships are associated with a statistically significant reduction in mortgage default probability of approximately 3.05% – 3.19% across the LPM, probit, and logit specifications for *Married Females*. This finding indicates that even among married women — who may benefit from spousal financial support — existing banking relationships provide meaningful incentives to avoid default.

Turning to marital and gender heterogeneity, the coefficient on the interaction term between *RelationshipDummy* and *Married Male* is negative and statistically significant at the 5% level across LPM, probit, and logit models. Economically, the relationship between prior banking relationships and the default probability for this group of borrowers are approximately 5.2% stronger relative to the benchmark group, i.e. *Married Female*, which is in line with our expectation. The coefficients on the interaction terms of *RelationshipDummy* and *Single Male* and *Single Female* are not statistically significant across different specification implying that the impact of prior relationships and the default probability is not significantly different for single female and single male borrowers relative to married female borrowers.

Overall, the results suggest that married male borrowers with prior relationship have stronger incentives — compared to other borrowers (single female, single male, and married female) — to fulfill their mortgage payment obligations, and thereby preserve their banking relationship.

8. Concluding Remarks

This study investigates the role of borrowers' prior banking relationship in loan repayments. Using a comprehensive dataset of around 149,230 mortgage loans, we find robust evidence that borrowers with prior banking relationships exhibit significantly lower propensity to default on their loan installments compared to those without such relationships. The results indicate that the borrowers' incentives channel plays a significant role in relationship banking.

Specifically, our LPM, probit, and logit estimates indicate that borrowers maintaining savings or current accounts prior to loan origination have an approximately 4% lower incentive to default, after controlling for an array of borrowers' characteristics and risk factors. This effect persists across various robustness

checks, including alternative estimation methods, different default definition, and measures of relationship duration, underscoring the pivotal role of relationship banking in mitigating default risk.

Furthermore, our analysis of the COVID-19 pandemic lockdown as an exogenous shock reveals that borrowers with established banking relationships are less inclined to default on their payment obligations, potentially driven by a desire to preserve valuable relationships that could facilitate future access to financial support. This finding highlights the stabilizing influence of relationship banking during economic crises. Similar results are also found when we consider the 2020 earthquake that struck West Azerbaijan and East Azerbaijan provinces. Overall, relationship banking appears to play a positive role during various types of crises and periods of hardship.

Our study also demonstrates the heterogeneity in borrowers' incentives to avoid default across different wealth levels. Using home prices as a proxy for wealth, we find that more affluent borrowers with prior banking relationships have greater incentives to honor their mortgage payments and thereby preserve their banking relationships possibly because they are more likely to apply for credit in the future.

In addition, our study further extends the literature by showing that borrowers' behavior under a mortgage loan varies across specific gender and marital status groups. We find that the association of prior banking relationships and incentives to avoid default is stronger for married male borrowers relative to other groups of borrowers. These findings highlight the importance of considering demographic factors in understanding the dynamics of borrowers' behavior.

In essence, this study underscores the significance of relationship banking as a powerful mechanism for aligning borrowers' incentives and enhancing loan repayment discipline. By nurturing long-term relationships, banks can cultivate a sense of commitment and trust among borrowers, ultimately reducing default risk and promoting financial stability. These insights carry significant implications for both lending institutions and policymakers, underscoring the importance of fostering enduring bank-borrower relationships as one of the fundamental aspects of a resilient financial system. Banks may consider assigning considerable importance to borrowers' incentives alongside screening and monitoring when assessing loan applications. Moreover, they may choose to prioritize extending loans to customers with longer-established relationships, recognizing the decreased default risk associated with such clients.

While our analysis relies on administrative loan-level data to identify and quantify borrowers' behavioral responses, future research could complement this approach with survey-based evidence to directly examine borrower motivations. In particular, collecting survey data on expectations regarding future credit access, perceived relationship benefits, or the costs and time required to establish new banking relationships would provide deeper insight into the mechanisms underlying the incentive effects documented in this paper. Such an approach could help disentangle whether repayment behavior is primarily driven by anticipated future financial advantages, switching costs, or broader reputational

concerns. We view this as a promising avenue for extending the current findings and further enriching our understanding of relationship-based incentives in credit markets.

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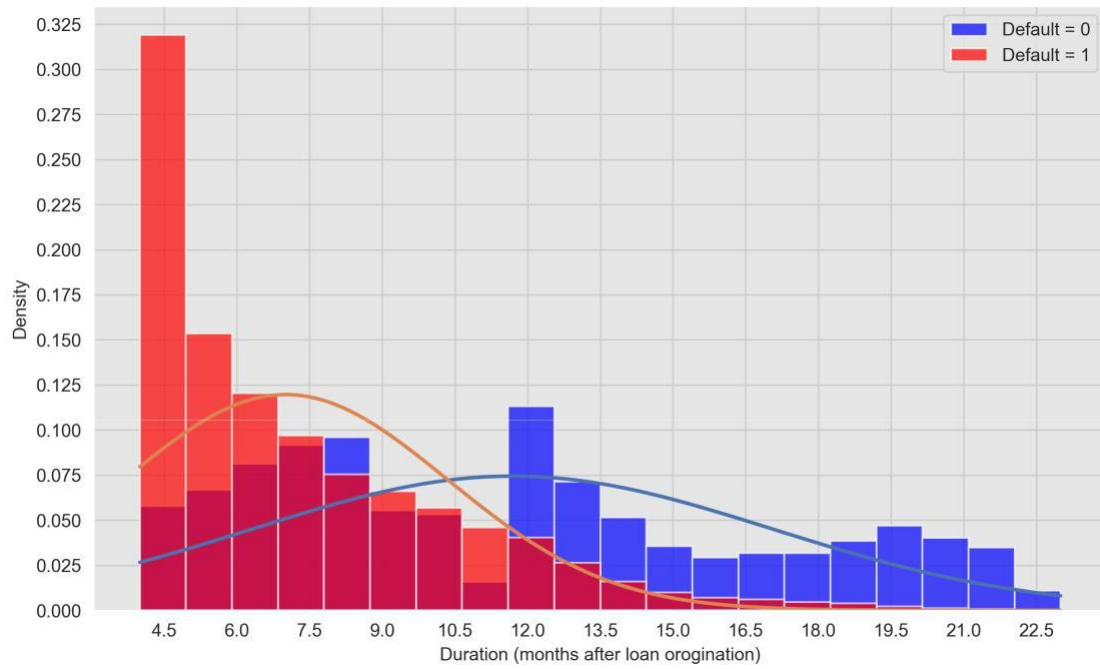
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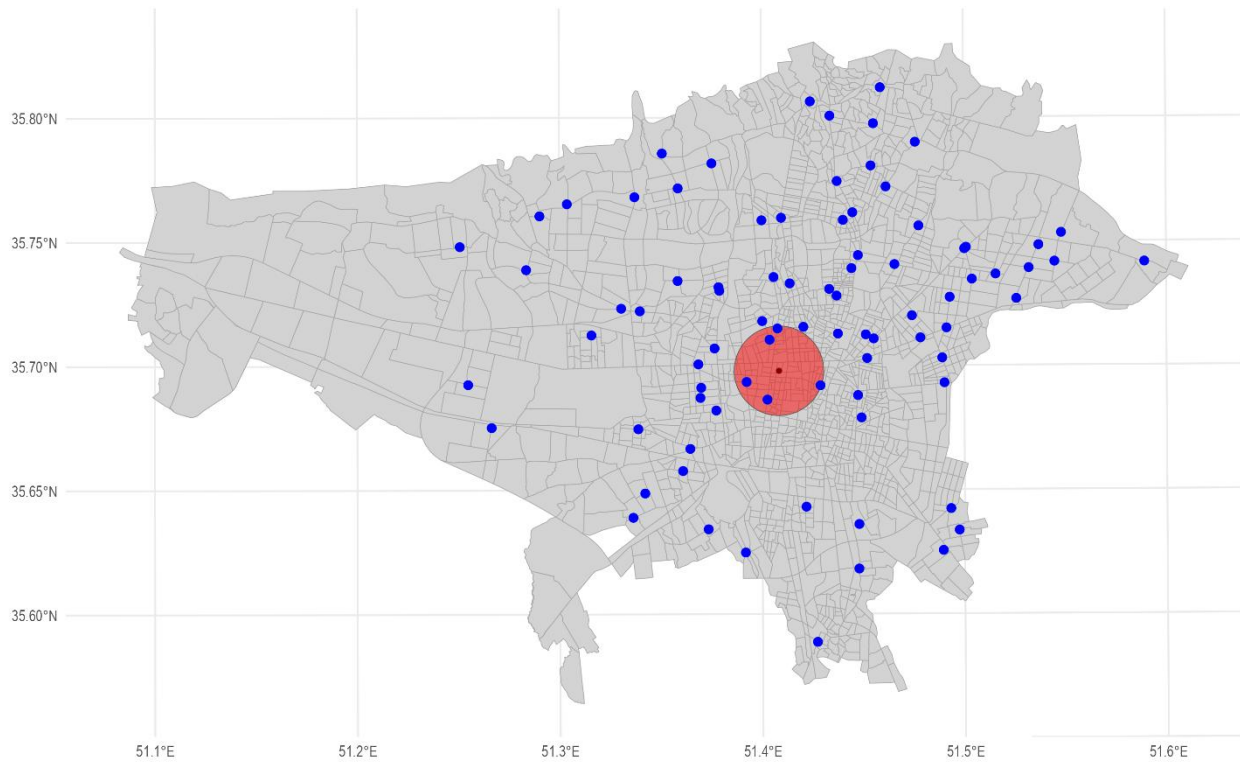
Figures

Figure 1: Histograms of Borrowers' Follow-Up Period



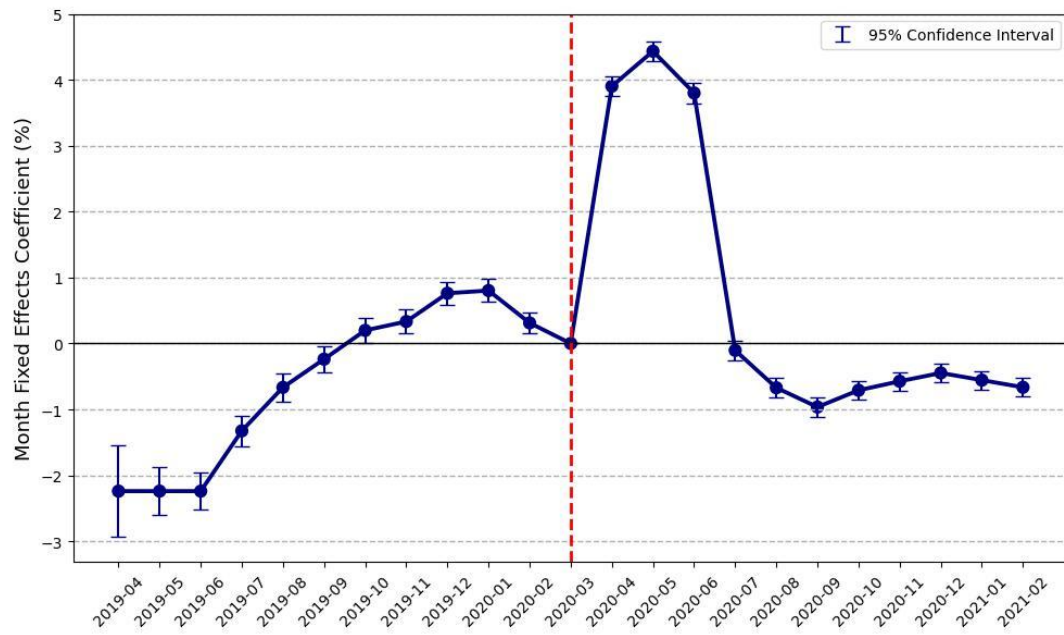
The figure depicts the histograms of the follow-up period for borrowers without default and those with default. The follow-up period is the duration (in months) that a borrower is observed in the data up until they exit the sample either as they default or the right-censoring. The x-axis represents the duration in months, while the y-axis represents the density of borrowers with a specific follow-up period.

Figure 2: Distribution of Maskan Bank Branches in Tehran, with a 2000m Radius Circle Around the Center of a Borrower's 5-Digit ZIP Code Neighborhood.



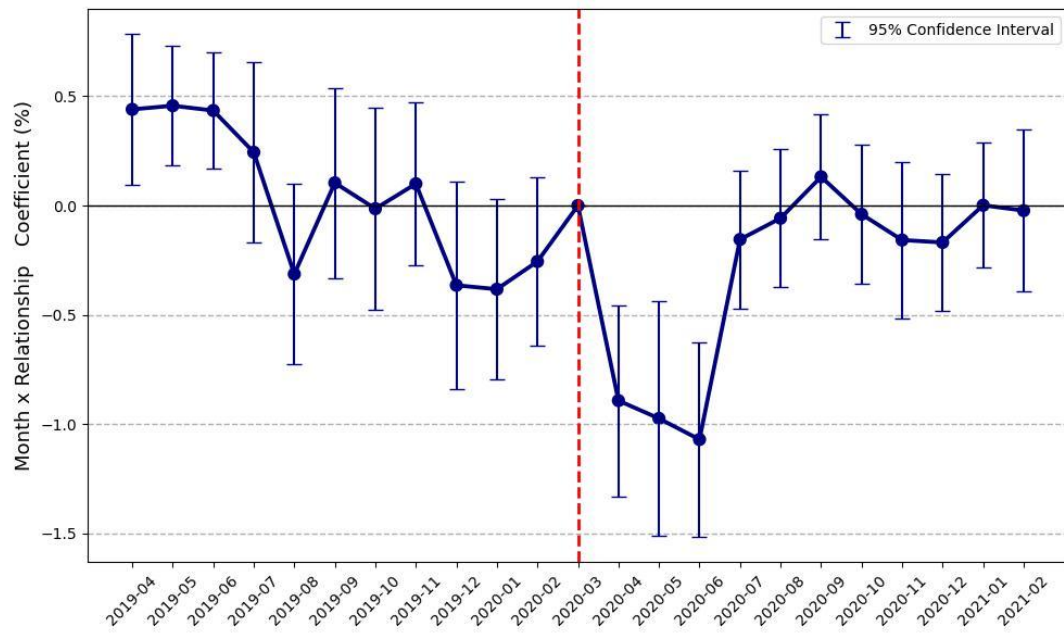
This figure illustrates the distribution of Maskan Bank branches across the map of Tehran. The red circle represents a 2000-meter radius, centered on a borrower's 5-digit ZIP code area. The x-axis depicts the longitude, while the y-axis displays the latitude.

Figure 3: COVID-19 Lockdown and the Loan Defaults Trend



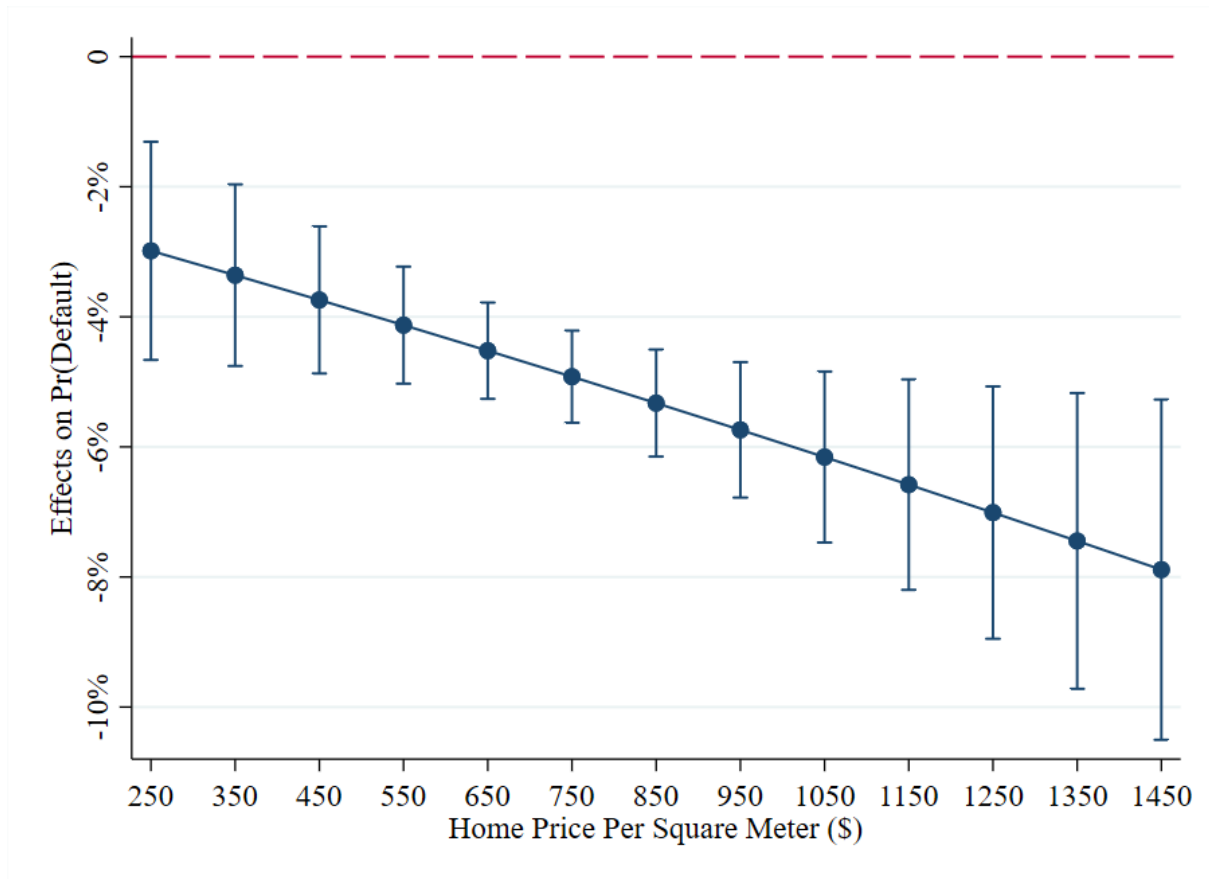
This figure shows the trend of loan defaults before and after the COVID-19 lockdown. The x-axis represents months, and the y-axis indicates the percentage of loan defaults. The vertical lines denote the 95% confidence interval. The benchmark month is March 2020.

Figure 4: The Effect of Prior Relationships in Loan Default During the COVID-19 Lockdown



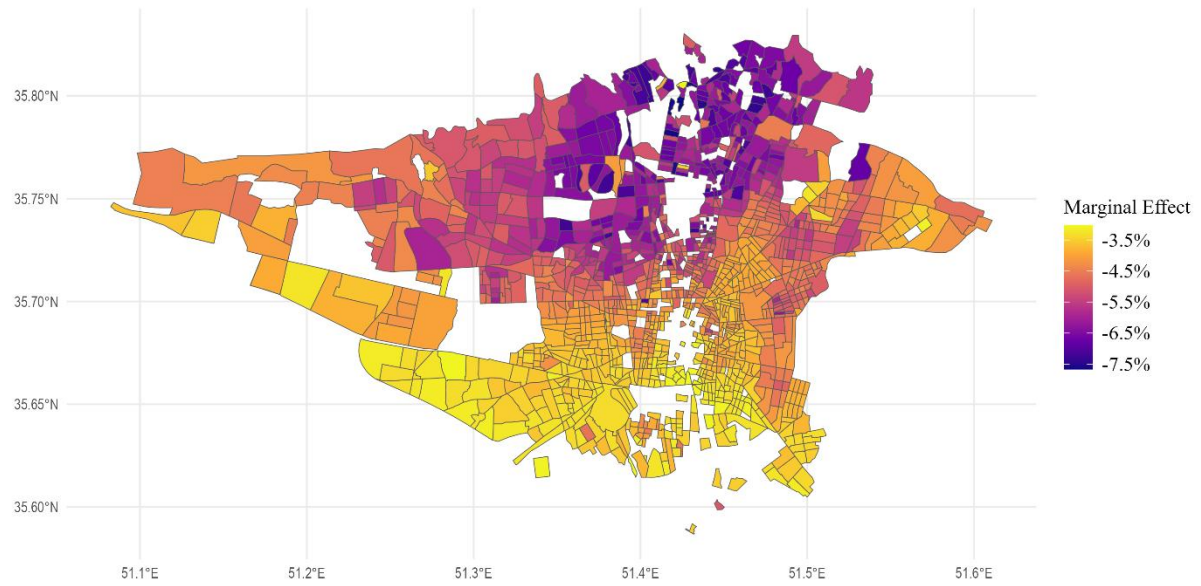
This figure depicts the relationship between a borrower's prior relationships with Maskan Bank and the risk of loan default during the COVID-19 lockdown period. The x-axis represents the month and the y-axis indicates the coefficients on *Month*×*RelationshipDummy* interaction term resulted from Equation (6). The vertical lines denote the %95 confidence interval. The benchmark month is March 2020.

Figure 5: Marginal Effect of Having Prior Relationships on Probability of Default across Different Home Price Brackets



This figure depicts the average marginal effect of having prior relationships with the bank on the probability of loan default across different home prices. The x-axis represents the home price per square meter in \$ and the y-axis indicates the change in the probability of default in percentage resulted from Equation (8). The vertical lines denote the 95% confidence interval.

Figure 6: Heterogeneous Effect of Having a Prior Relationship on Probability of Default across Tehran's Zones



This figure depicts the average marginal effect (%) of having prior relationships on the probability of default at each district zone of Tehran. The x-axis represents longitude and the y-axis indicates latitude. The color of each zone reflects a specified marginal effect in percentage, indicated in the legend bar. The districts without color represent missing values.

Tables

Table 1: Summary Statistics

	Mean	SD	Min	Max	N
3-Month Default (%)	26.49	44.12	0.00	100	149,230
6-Month Default (%)	5.82	23.41	0.00	100	131,784
RelationshipDummy (%)	34.25	47.45	0.00	100	149,230
RelationshipDuration (in Years)	9.33	5.78	0.00	30.66	51,121
Male (%)	58.37	49.29	0.00	100	149,230
Married (%)	86.71	33.94	0.00	100	149,230
Effective Rate (%)	20.42	1.08	17.92	22.97	149,230
Loan to Value (%)	34.56	21.14	4.13	83.33	146,291
Bounced Check Record (%)	3.65	18.76	0.00	100	149,230
Default Record in Previous Loans (%)	12.08	32.59	0.00	100	149,230
Loan Origination Time	8.03	7.26	0	31	149,230

This table presents the summary statistics of borrowers' characteristics, risk measures, and loan defaults. The sample includes loans originated by Maskan Bank under the *Second Program* of loan issuance from March 2019 to February 2021. Please refer to Appendix A3 for variable definitions.

Table 2: Covariate Balance Test

	Without Prior Relationship		With Prior Relationship		Overlap Measures			
	(N _{nr} = 96,438)		(N _r = 49,853)		Nor Dif	Log Ratio of STD	$\pi^{0.05}$	
	Mean	(S.D.)	Mean	(S.D.)			Without Prior Relationship	With Prior Relationship
Male	0.577	(0.494)	0.596	(0.491)	0.040	-0.007	0	0
Married	0.884	(0.321)	0.836	(0.371)	-0.139	0.145	0	0
Loan to Value	0.357	(0.211)	0.323	(0.209)	-0.162	-0.014	0.018	0.042
Bounced Check	0.028	(0.166)	0.050	(0.218)	0.112	0.274	0	0
Default in Previous Loans	0.122	(0.327)	0.120	(0.325)	-0.006	-0.007	0	0
Effective Rate	0.2041	(0.0107)	0.2042	(0.0108)	0.008	0.007	0.045	0.051
Loan Origination Time	7.971	(7.27)	8.066	(7.243)	0.013	-0.004	0.019	0.019
Multivariate Measures								
P-Score	0.336	(0.052)	0.350	(0.058)	0.242	0.097	0.047	0.064
Linearized P-Score	-0.688	(0.233)	-0.630	(0.253)	0.241	0.081	0.047	0.064

This table presents the overlap assessment of covariates between borrowers with and without prior relationships with the bank. The overlap measures are normalized differences (*Nor Dif*), dispersion difference (*Log Ratio of STD*), and coverage frequency ($\pi^{0.05}$). Additionally, multivariate measures such as propensity scores (*P-Score*) and linearized propensity scores (*Linearized P-Score*) are reported to assess the overall balance between the two groups. N_{nr} and N_r represent the number of borrowers with and without relationship.

Table 3: Average Marginal Effects of LPM, Probit, and Logit Regressions

	(1)	(2)	(3)	(4)
	LPM	Probit	Logit	LPM
<i>Borrower's Relationship</i>				
RelationshipDummy	-0.0403*** (0.0024)	-0.0404*** (0.0024)	-0.0408*** (0.0024)	-0.0369*** (0.0024)
<i>Borrower's Characteristics</i>				
Male	-0.0017 (0.0022)	-0.0011 (0.0023)	-0.0014 (0.0023)	-0.0020 (0.0023)
Married	-0.0002 (0.0036)	0.0004 (0.0037)	0.0001 (0.0036)	-0.0015 (0.0037)
<i>Borrower's Risk Measures</i>				
Loan to Value	0.0007*** (0.0001)	0.0007*** (0.0001)	0.0007*** (0.0001)	0.0007*** (0.0001)
Bounced Check	0.1549*** (0.0102)	0.1372*** (0.0086)	0.1384*** (0.0085)	0.1486*** (0.0101)
Default in Previous Loans	0.4958*** (0.0042)	0.3826*** (0.0029)	0.3719*** (0.0029)	0.5028*** (0.0042)
Effective Rate	0.0099*** (0.0015)	0.0111*** (0.0018)	0.0119*** (0.0018)	0.0084*** (0.0015)
Job	Yes	Yes	Yes	Yes
Education	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes
City	No	No	No	Yes
N	146,291	146,291	146,291	144,323
R ² (Pseudo R ²)	0.2294	0.1987	0.1986	0.2434
F (Wald Chi2)	569.87	18378.96	15498.05	618.77

This table exhibits the average marginal effects resulted from Equation (1) using the LPM, probit, and logit estimation. The dependent variable is a binary variable equal to 1 if the borrower defaults within the full observation period after loan origination where the default is defined as failure to repay the installments for three consecutive months. The variable of interest is *RelationshipDummy*. We control for *Male*, *Married*, *Loan to Value*, *Bounced Check*, *Default in Previous Loans*, and *Effective Rate*. We also include *Job*, *Education*, *Month*, and *City* fixed effects, while we do not report their coefficients for the sake of brevity. The error terms are clustered two-way at the Loan Origination Month and City levels in all models. The standard errors are calculated using the Delta method. Please refer to Appendix A3 for variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Table 4: Descriptive Statistics of Censored and Failed Subjects

Subject Type	Mean	SD	Min	Max	N
Censored Subjects (Default = 0)	11.60	5.40	3	23	109,706
Failed Subjects (Default = 1)	6.75	3.37	3	23	39,524

This table presents the statistics on censored and failed borrowers within the survival data. A censored borrower refers to one who exits the dataset for reasons other than default. Conversely, a failed borrower is one who exits the dataset due to default.

Table 5: Cox Proportional Hazard Regression

	(1) Hazard Ratio	(2) Hazard Ratio	(3) Hazard Ratio
<i>Borrower's Relationship</i>			
RelationshipDummy	0.818*** (0.0100)	0.844*** (0.0104)	0.807*** (0.0100)
<i>Borrower's Characteristics</i>			
Male		1.105*** (0.0129)	0.982 (0.0112)
Married		1.074** (0.0206)	0.987 (0.0186)
<i>Borrower's Risk Measures</i>			
Loan to Value			1.003*** (0.0002)
Bounced Check			1.745*** (0.0618)
Default in Previous Loans			4.856*** (0.0775)
Effective Rate			1.071*** (0.0098)
Job	No	Yes	Yes
Education	No	Yes	Yes
Month	No	Yes	Yes
N	149,230	149,230	146,291
Wald Chi2	269.56	1874.04	16611.81

This table exhibits the hazard ratios obtained from the Cox proportional hazard model (Equation (2)). The dependent variable is the time to 3-month default indicator. Column (1) includes the *RelationshipDummy* as the explanatory variable. In column (2), we add borrowers' characteristics, i.e. *Male* and *Married*. Column (3) reports the result when we incorporate borrowers' risk measures, i.e., *Loan to Value*, *Bounced Check*, *Default in Previous Loans*, and *Effective Rate*. We also include *Job*, *Education*, and *Month* fixed effects in columns (2) and (3), while we do not report their coefficients for the sake of brevity. The error terms are clustered two-way at the Loan Origination Month and City levels in all models. Standard errors of exponentiated coefficients are in parenthesis. Please refer to Appendix A3 for variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Table 6: Default Probability Prediction by Machine Learning Models

	(1) Random Forest	(2) XGBoost	(3) Neural Networks
<i>Borrower's Relationship</i>			
RelationshipDummy	-0.0323*** (0.0024)	-0.0395*** (0.0028)	-0.0380*** (0.0021)
<i>Borrower's Characteristics</i>			
Male	0.0011 (0.0030)	0.0088*** (0.0015)	0.0110*** (0.0032)
Married	0.0091** (0.0035)	0.0161*** (0.0017)	0.0135*** (0.0035)
<i>Borrower's Risk Measures</i>			
Loan to Value	-0.0003 (0.0021)	-0.0086** (0.0035)	0.0287*** (0.0026)
Bounced Check	0.1639*** (0.0014)	0.2194*** (0.0019)	0.2417*** (0.0033)
Default in Previous Loans	0.4266*** (0.0023)	0.5140*** (0.0018)	0.5323*** (0.0027)
Effective Rate	-0.0869*** (0.0018)	-0.1374*** (0.0035)	-0.2634*** (0.0026)
Job	Yes	Yes	Yes
Education	Yes	Yes	Yes
ROC AUC	0.7316	0.7603	0.7507
Average Precision Score	0.5479	0.5765	0.5663
Brier Score	0.1581	0.1518	0.1526
R ²	0.1795	0.2121	0.2080

This table shows the effect of variables on the probability of default using three machine learning models: Random Forest, XGBoost, and Neural Networks, as presented in columns (1), (2), and (3), respectively. The outcome variable is binary, representing a classification problem where the outcome is one if the borrower defaults within the full observation period after loan origination. Default is defined as failing to repay installments for three consecutive months. The variable of interest is *RelationshipDummy*. We control for *Male*, *Married*, *Loan to Value*, *Bounced Check*, *Default in Previous Loans*, and *Effective Rate*. We also include *Job* and *Education* as categorical features; however, for brevity, we do not report their effects. The standard errors are computed using bootstrap sampling. Please refer to Appendix A3 for variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Table 7: Summary Statistics of the Number of Maskan Bank Branches and Branches of other Banks Inside the Circles with Radii of 500, 1000, 1500, 2000 Meters.

	Mean	SD	Min	Max	N
<i>Maskan Bank Branches</i>					
<i>500m Circle</i>	0.20	0.43	0.00	2.00	28,430
<i>1000m Circle</i>	0.71	0.80	0.00	4.00	28,430
<i>1500m Circle</i>	1.41	1.22	0.00	6.00	28,430
<i>2000m Circle</i>	2.38	1.72	0.00	8.00	28,430
<i>Other Banks Branches</i>					
<i>500m Circle</i>	2.97	4.11	0.00	32.00	28,430
<i>1000m Circle</i>	11.19	11.73	0.00	72.00	28,430
<i>1500m Circle</i>	24.04	22.70	0.00	141.00	28,430
<i>2000m Circle</i>	41.56	37.10	0.00	230.00	28,430

This table presents summary statistics for the number of Maskan Bank branches and other banks located within circles of varying radii — 500, 1000, 1500, and 2000 meters — centered at the borrowers' 5-digit ZIP code neighborhood center.

Table 8: The Role of Geographic Distance to Maskan Bank Branches

	(1)	(2)	(3)	(4)
	500m Circle	1000m Circle	1500m Circle	2000m Circle
Number of Maskan Branches (<i>NMB</i>)	0.0157** (0.00675)	0.00632* (0.00369)	0.00759*** (0.00253)	0.00781*** (0.00183)
Male	0.000832 (0.00611)	0.000840 (0.00611)	0.000872 (0.00611)	0.00107 (0.00611)
Married	-0.0374*** (0.00756)	-0.0374*** (0.00756)	-0.0370*** (0.00757)	-0.0370*** (0.00756)
Bounced Check	0.107*** (0.0195)	0.107*** (0.0195)	0.108*** (0.0195)	0.109*** (0.0195)
Default in Previous Loans	0.0562*** (0.0155)	0.0562*** (0.0155)	0.0565*** (0.0155)	0.0571*** (0.0155)
Job	Yes	Yes	Yes	Yes
Education	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes
District	Yes	Yes	Yes	Yes
N	27,233	27,233	27,233	27,233
Pseudo R ²	0.0137	0.0137	0.0138	0.0141
Wald Chi2	477.12	474.72	480.67	489.64

This table presents the marginal effects of the number of Maskan Bank branches located inside circles with varying radii — 500, 1000, 1500, and 2000 meters — centered at the neighborhood center of borrowers' 5-digit ZIP codes on having relationship with Maskan Bank. The dependent variable is a binary variable that takes the value of one if the borrower has a relationship with Maskan Bank and zero otherwise. The variable of interest is *Number of Maskan Branches (NMB)*. We control for *Male*, *Married*, *Bounced Check*, and *Default in Previous Loans*. We also include *Job*, *Education* and *Month* fixed effects, while we do not report their coefficients for the sake of brevity. Robust standard errors are in parentheses. Please refer to Appendix A3 for variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Table 9: The Marginal Effects of Distance to Maskan Branch (MBD) on Having Relationship with Maskan Bank

	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
	Nearest Branch	2 Nearest Branches	3 Nearest Branches	4 Nearest Branches	5 Nearest Branches	6 Nearest Branches	7 Nearest Branches	8 Nearest Branches	9 Nearest Branches	10 Nearest Branches
Distance to Maskan Branch (MBD)	-0.00185*** (0.000301)	-0.00211*** (0.000305)	-0.00215*** (0.000294)	-0.00207*** (0.000280)	-0.00197*** (0.000267)	-0.00190*** (0.000253)	-0.00184*** (0.000242)	-0.00180*** (0.000232)	-0.00176*** (0.000223)	-0.00174*** (0.000217)
Male	0.000482 (0.00611)	0.000718 (0.00611)	0.000858 (0.00611)	0.000961 (0.00611)	0.000986 (0.00611)	0.00104 (0.00611)	0.00111 (0.00611)	0.00118 (0.00611)	0.00123 (0.00611)	0.00127 (0.00611)
Married	-0.0363*** (0.00757)	-0.0360*** (0.00756)	-0.0358*** (0.00756)	-0.0358*** (0.00756)	-0.0358*** (0.00756)	-0.0357*** (0.00756)	-0.0357*** (0.00756)	-0.0357*** (0.00756)	-0.0357*** (0.00756)	-0.0357*** (0.00756)
Bounced Check	0.103*** (0.0195)	0.104*** (0.0196)	0.105*** (0.0196)	0.105*** (0.0196)	0.106*** (0.0196)	0.106*** (0.0196)	0.106*** (0.0196)	0.106*** (0.0196)	0.106*** (0.0196)	0.106*** (0.0196)
Default in Previous Loans	0.0532*** (0.0154)	0.0534*** (0.0155)	0.0538*** (0.0155)	0.0540*** (0.0155)	0.0540*** (0.0155)	0.0540*** (0.0155)	0.0540*** (0.0155)	0.0540*** (0.0155)	0.0540*** (0.0155)	0.0540*** (0.0155)
Job	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	27,233	27,233	27,233	27,233	27,233	27,233	27,233	27,233	27,233	27,233
Pseudo R ²	0.0152	0.0156	0.0158	0.0158	0.0158	0.0158	0.0159	0.0159	0.0160	0.0160
Wald Chi ²	527.42	536.56	541.52	543.09	543.39	545.00	546.92	548.78	550.68	552.68

This table presents the marginal effects of the average distance between the borrowers' 5-digit ZIP code neighborhood center and k nearest Maskan Bank branches on having relationship with Maskan Bank. The dependent variable is a binary variable that takes the value of one if the borrower has a relationship with Maskan Bank and zero otherwise. The variable of interest is *Distance to Maskan Bank Branches (MBD)*. We control for *Male*, *Married*, *Bounced Check*, and *Default in Previous Loans*. We also include *Job*, *Education* and *Month* fixed effects, while we do not report their coefficients for the sake of brevity. Robust standard errors are in parentheses. Please refer to Appendix A3 for variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Table 10: The Marginal Effects of Distance Ratio (DR) on Having Relationship with Maskan Bank

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k = 7$	$k = 8$	$k = 9$	$k = 10$
Distance Ratio (DR)	-0.00230*** (0.000880)	-0.00419*** (0.00124)	-0.00487*** (0.00148)	-0.00572*** (0.00171)	-0.00637*** (0.00191)	-0.00691*** (0.00210)	-0.00756*** (0.00226)	-0.00843*** (0.00241)	-0.00935*** (0.00258)	-0.0101*** (0.00273)
Male	0.00114 (0.00611)	0.00104 (0.00611)	0.00103 (0.00611)	0.00108 (0.00611)	0.00112 (0.00611)	0.00114 (0.00611)	0.00114 (0.00611)	0.00114 (0.00611)	0.00113 (0.00611)	0.00112 (0.00611)
Married	-0.0371*** (0.00757)	-0.0369*** (0.00757)	-0.0369*** (0.00757)	-0.0369*** (0.00757)	-0.0369*** (0.00757)	-0.0369*** (0.00757)	-0.0369*** (0.00757)	-0.0369*** (0.00757)	-0.0369*** (0.00757)	-0.0369*** (0.00757)
Bounced Check	0.108*** (0.0195)	0.108*** (0.0195)	0.108*** (0.0195)	0.108*** (0.0195)	0.108*** (0.0195)	0.108*** (0.0195)	0.108*** (0.0195)	0.108*** (0.0195)	0.108*** (0.0195)	0.108*** (0.0195)
Default in Previous Loans	0.0544*** (0.0154)	0.0547*** (0.0154)	0.0548*** (0.0154)	0.0550*** (0.0154)	0.0550*** (0.0154)	0.0550*** (0.0154)	0.0550*** (0.0154)	0.0551*** (0.0154)	0.0551*** (0.0154)	0.0550*** (0.0154)
Job	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	27,233	27,233	27,233	27,233	27,233	27,233	27,233	27,233	27,233	27,233
Pseudo R ²	0.0152	0.0152	0.0152	0.0152	0.0152	0.0152	0.0152	0.0152	0.0153	0.0153
Wald Chi ²	0.0151	531.73	531.38	531.42	531.40	531.21	531.45	532.46	533.26	533.75

This table presents the marginal effects of the *Distance Ratio (DR)* on having relationship with Maskan Bank. The dependent variable is a binary variable that takes the value of one if the borrower has a relationship with Maskan Bank and zero otherwise. The variable of interest is the distance ratio (*DR*). We control for *Male*, *Married*, *Bounced Check*, and *Default in Previous Loans*. We also include *Job*, *Education* and *Month* fixed effects, while we do not report their coefficients for the sake of brevity. Robust standard errors are in parentheses. Please refer to Appendix A3 for variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Table 11: Average Marginal Effects of LPM, Probit, and Logit Regressions – Controlling for Borrowers' Wealth

	(1)	(2)	(3)
	LPM	Probit	Logit
<i>Borrower's Relationship</i>			
RelationshipDummy	-0.0424*** (0.00503)	-0.0427*** (0.00497)	-0.0431*** (0.00508)
<i>Borrower's Characteristics</i>			
Male	0.00423 (0.00464)	0.00482 (0.00461)	0.00442 (0.00466)
Married	0.0129** (0.00570)	0.0150** (0.00591)	0.0137** (0.00592)
Home Price	0.0345*** (0.0123)	0.0328*** (0.0118)	0.0332*** (0.0121)
<i>Borrower's Risk Measures</i>			
Loan to Value	0.000973*** (0.000102)	0.000938*** (0.0000916)	0.000938*** (0.0000912)
Bounced Check	0.130*** (0.0310)	0.112*** (0.0231)	0.108*** (0.0222)
Default in Previous Loans	0.437*** (0.0156)	0.300*** (0.00987)	0.287*** (0.00965)
Effective Rate	0.00754** (0.00249)	0.00845*** (0.00280)	0.00880*** (0.00289)
Job	Yes	Yes	Yes
Education	Yes	Yes	Yes
Month	Yes	Yes	Yes
District	Yes	Yes	Yes
N	26,581	26,560	26,560
R ² (Pseudo R ²)	0.127	0.132	0.131

This table exhibits the average marginal effects resulted from estimation of Equation (1) using the LPM, probit, and logit estimation techniques when we include *Home Price* as a control variable. The sample is limited to Tehran due to data limitation. The dependent variable is a binary variable that takes the value of one if the borrower defaults within the full observation period after loan origination where the default is defined as failure to repay the installments for three consecutive months. The variable of interest is *RelationshipDummy*. We control for *Male*, *Married*, *Home Price*, *Loan to Value*, *Bounced Check*, *Default in Previous Loans*, and *Effective Rate*. We also include *Job*, *Education*, *Month* and *District* fixed effects, while we do not report their coefficients for the sake of brevity. The error terms are clustered two-way at the Loan Origination Month and District levels in all models. The standard errors are calculated using the Delta method. Please refer to Appendix A3 for variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Table 12: Covariate Balance Test – Treatment and Control Samples

	Unaffected Counties		Affected Counties		Overlap Measures			
	(N _c = 5,856)		(N _t = 4,659)		Nor Dif	Log Ratio of STD	$\pi^{0.05}$	
	Mean	(S.D.)	Mean	(S.D.)			Unaffected Counties	Affected Counties
RelationshipDummy	0.344	(0.475)	0.336	(0.472)	-0.016	-0.006	0	0
Male	0.627	(0.483)	0.625	(0.484)	-0.006	0.002	0	0
Married	0.872	(0.334)	0.869	(0.338)	-0.010	0.011	0	0
Loan to Value	0.347	(0.212)	0.341	(0.209)	-0.034	-0.014	0.035	0.033
Bounced Check	0.031	(0.174)	0.040	(0.197)	0.049	0.124	0	0
Default in Previous Loans	0.168	(0.374)	0.181	(0.385)	0.033	0.029	0	0
Effective Rate	0.1983	(0.0062)	0.1980	(0.0063)	-0.054	0.016	0.050	0.051
Multivariate Measures								
P-Score	0.454	(0.022)	0.456	(0.022)	0.089	0.022	0.050	0.050
Linearized P-Score	-0.186	(0.089)	-0.178	(0.09)	0.090	0.021	0.050	0.050

This table presents the overlap assessment of covariates between borrowers with and without prior relationships with the bank. The overlap measures are normalized differences (*Nor Dif*), dispersion difference (*Log Ratio of STD*), and coverage frequency ($\pi^{0.05}$). Additionally, multivariate measures such as propensity scores (*P-Score*) and linearized propensity scores (*Linearized P-Score*) are reported to assess the overall balance between the two groups. N_c and N_t represent the number of borrowers in control and treatment groups.

Table 13: The Marginal Effects of Triple Difference Term

	(1) LPM	(2) Probit	(3) Logit
2019-09 × Quake × RelationshipDummy	0.00032 (0.00906)	0.00213 (0.00818)	0.00146 (0.00812)
2019-10 × Quake × RelationshipDummy	0.00987 (0.01311)	0.00672 (0.01186)	0.0092 (0.01237)
2019-11 × Quake × RelationshipDummy	-0.00855 (0.00967)	-0.01002 (0.00887)	-0.00788 (0.00882)
2019-12 × Quake × RelationshipDummy	0.01000 (0.01589)	0.00656 (0.01470)	0.01003 (0.01486)
2020-01 × Quake × RelationshipDummy	0.00880 (0.01650)	0.00616 (0.01343)	0.00793 (0.01570)
2020-03 × Quake × RelationshipDummy	-0.01314* (0.0074)	-0.01562** (0.00750)	-0.01291* (0.00747)
2020-04 × Quake × RelationshipDummy	0.00952 (0.01272)	0.00700 (0.01184)	0.00763 (0.01270)
2020-05 × Quake × RelationshipDummy	0.01986 (0.01454)	0.01631 (0.01428)	0.01860 (0.01400)
2020-06 × Quake × RelationshipDummy	0.00434 (0.01121)	0.00389 (0.01142)	0.00297 (0.01190)
2020-07 × Quake × RelationshipDummy	0.00130 (0.00890)	0.00120 (0.00935)	0.00169 (0.00939)
RelationshipDummy	Yes	Yes	Yes
Earthquake	Yes	Yes	Yes
Month	Yes	Yes	Yes
Quake × RelationshipDummy	Yes	Yes	Yes
Quake × Month	Yes	Yes	Yes
Month × RelationshipDummy	Yes	Yes	Yes
Characteristics	Yes	Yes	Yes
RiskMeasures	Yes	Yes	Yes
R ² (Pseudo R ²)	0.0370	0.0901	0.0901
F (Wald Chi2)	19694.52	1.15e+08	2.67e+08
N	63,394	63,394	63,394

This table exhibits the average marginal effects resulted from Equation (7) using the LPM, probit, and logit. The study period is from September 2019 to July 2020. The benchmark month is February 2020. The dependent variable is a binary variable that takes the value of one if the borrower defaults within the full observation period after loan origination, where the default is defined as failure to repay the installments for three consecutive months. We include *RelationshipDummy*, *Earthquake*, *Month*, and their double interactions. We also control for borrower's *Characteristics* and *RiskMeasures*; however, we do not report their coefficients for the sake of brevity. Standard errors are clustered at the city level and calculated using the Delta method. Please refer to Appendix A3 variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Table 14: The Marginal Effects of Prior Relationship – The Role of Gender and Marital Status

	(1) LPM	(2) Probit	(3) Logit
RelationshipDummy	-0.0305*** (0.0121)	-0.0319*** (0.0091)	-0.0315*** (0.0092)
RelationshipDummy × Single Male	-0.0186 (0.0215)	-0.0166 (0.0213)	-0.0206 (0.0239)
RelationshipDummy × Married Male	-0.0214** (0.0098)	-0.0197** (0.0100)	-0.0205** (0.0101)
RelationshipDummy × Single Female	0.0044 (0.0140)	0.0065 (0.0153)	0.0029 (0.0154)
Single Male	0.0095 (0.0122)	0.0061 (0.0117)	0.0077 (0.0115)
Married Male	0.0034 (0.0053)	0.0038 (0.0051)	0.0029 (0.0052)
Single Female	-0.0411*** (0.0091)	-0.0453*** (0.0115)	-0.0449*** (0.0120)
Loan to Value	0.000977*** (0.000138)	0.000941*** (0.000112)	0.000943*** (0.000111)
Bounced Check	0.129*** (0.0233)	0.112*** (0.0169)	0.108*** (0.0168)
Default in Previous Loans	0.438*** (0.0165)	0.301*** (0.0106)	0.288*** (0.0103)
Effective Rate	0.00754** (0.00315)	0.00841** (0.00350)	0.00882** (0.00361)
Home Price	0.0343*** (0.0126)	0.0329*** (0.0123)	0.0332*** (0.0122)
Job	Yes	Yes	Yes
Education	Yes	Yes	Yes
Month	Yes	Yes	Yes
District	Yes	Yes	Yes
N	26,581	26,560	26,560
R ² (Pseudo R ²)	0.1277	0.1308	0.1302
F (Wald Chi2)	78.66	4797.90	4170.80

This table exhibits the average marginal effects resulted from Equation (10) using the LPM, probit, and logit estimations. The dependent variable is a binary variable that takes the value of one if the borrower defaults within the full observation period after loan origination, where the default is defined as failure to repay the installments for three consecutive months. The variable of interest is the interaction term between between the *RelationshipDummy* and borrower group dummies — *Single Male*, *Married Male*, and *Married Female* — using *Single Female* borrowers as the reference category. We control for *Neighborhood Home Price*, *Loan to Value*, *Bounced Check*, *Default in Previous Loans*, *Effective Rate*. We also include *Job*, *Education*, *Month*, and *District* fixed effects, while we do not report their coefficients for the sake of brevity. The error terms are clustered two-way at the Loan Origination Month and District levels in all models. Please refer to Appendix A3 variable definitions. ***, ** and * indicate significance at 1%, 5% and 10%, respectively.

Appendices

Appendix A1. Theoretical Framework

This section presents an intertemporal consumption-smoothing model to explore how borrower-lender relationship can create incentives for borrowers to reduce default behavior. The framework captures an individual's consumption and saving decisions over three periods, allowing for borrowing and partially unpaid debt. The pivotal component of the model is the introduction of a time-varying borrower-lender relationship capital k_t , which evolves endogenously based on repayment history and directly affects borrowing costs. We show that stronger banking relationships — reflected in higher values of k_t — lead to lower default rates by reducing borrowing costs and increasing the long-term value of maintaining the borrower-lender relationship.

A1.1. Model Setup

We develop a simple three-period model in which a representative individual maximizes lifetime utility. In periods 1 and 2, she chooses the levels of borrowing (or savings), denoted b_1 and b_2 , and in period 2 she also selects the amount of unpaid debt u_1 . The individual receives exogenous incomes y_1 , y_2 , and y_3 in each period. During the first two periods, she can borrow or save at interest rates r_1 and r_2 , respectively. If she borrows, she may strategically default on part of her obligation by leaving an amount u_1 unpaid: a larger u_1 signifies weaker repayment performance. Given these choices, consumption in each period is determined as follows:

$$\text{Period 1: } c_1 = y_1 + b_1 ,$$

$$\text{Period 2: } c_2 = y_2 + b_2 - (1 + r_1) b_1 + u_1 ,$$

$$\text{Period 3: } c_3 = y_3 - (1 + r_2) b_2 - (1 + \gamma) u_1 .$$

Where $\gamma > 0$ is the penalty rate applied to any unpaid balance carried forward.

The individual's choices are subject to the following constraints:

$$b_t \leq \theta \sum_{\tau=t}^T \beta^{\tau-t} y_\tau \quad (\text{Borrowing limit})$$

$$0 \leq u_t \leq (1 + r_t) b_t \quad (\text{Unpaid debt ceiling})$$

Similar to [Dobkin et al. \(2018\)](#), we define borrowing limit as a fraction ($0 < \theta \leq 1$) of the present discounted value of future income, where β is the intertemporal discount factor. The unpaid debt ceiling ensures that default cannot exceed the total repayment obligation in a given period.

The model assumes the following initial conditions:

$$b_0 = 0, u_{-1} = 0.$$

To ensure a closed-form solution, we impose terminal conditions requiring that all outstanding debts are fully paid by the end of the final period:

$$b_3 = 0 \quad (\text{No borrowing in the final period})$$

$$u_2 = 0 \quad (\text{All outstanding debt must be repaid by the final period})$$

For the sake of simplicity, borrowing costs are modeled solely through the interest rate $r_t(b_t, k_t)$. The interest rate is assumed to increase with the volume of borrowing, i.e., $\partial r_t / \partial b_t > 0$, and decrease with the strength of the borrower-lender relationship, $\partial r_t / \partial k_t < 0$. This structure captures the intertemporal trade-offs borrowers face between current consumption and future access to credit. Furthermore, the interest rate is assumed to be convex in relationship strength, implying that the marginal benefit of improving one's relationship with the bank (lender) — measured in terms of reduced interest rates — diminishes as the relationship is stronger ($\partial^2 r_t / \partial k_t^2 > 0$). These assumptions are broadly consistent with findings in the relationship lending literature (Petersen and Rajan, 1994; Berger and Udell, 1995; Boot and Thakor, 2000; Degryse and Van Cayseele, 2000). Therefore, we define the interest rate as:

$$r_t = \begin{cases} r_a + r_t(b_t, k_t) & \text{if } b_t > 0, \\ r_a & \text{if } b_t \leq 0. \end{cases} \quad (\text{A1.1})$$

Where r_a is the interest rate on savings. The borrowing interest rate exceeds r_a and is determined by the amount of borrowing b_t and the borrower-lender relationship capital k_t .

The borrower-lender relationship, denoted k_t , evolves endogenously as a function of past loan repayment behavior. Specifically, higher levels of unpaid debt in the previous period weaken the relationship, while lower levels help strengthen it. Furthermore, the negative impact of default is assumed to be more pronounced for borrowers with initially stronger banking relationships. This assumption is supported by existing literature (Diamond, 1989; Petersen and Rajan, 1994; Berger and Udell, 1995; Boot and Thakor, 2000; Degryse and Van Cayseele, 2000).

We introduce a threshold level of unpaid debt, denoted $\bar{u}$, such that when the amount of unpaid debt exceeds this threshold, the borrower-lender relationship deteriorates. The extent to which unpaid debt exceeds the threshold reflects the borrower's effective default intensity. In contrast, unpaid debt below the threshold does not harm the relationship. Improved repayment behavior — reflected in lower levels of unpaid debt below the threshold — can enhance the borrower's relationship with the bank.

The dynamic evolution of the borrower-lender relationship is captured by the following function:

$$k_t = f(k_{t-1}, u_{t-1}) \quad \text{with initial condition } k_1 = \bar{k}, \quad (\text{A1.2})$$

Subject to the following properties:

$$\frac{\partial k_t}{\partial u_{t-1}} < 0, \quad \frac{\partial^2 k_t}{\partial u_{t-1}^2} > 0$$

That is, the relationship declines with higher unpaid debt, and the marginal deterioration is decreasing in the level of default.

Finally, the individual maximizes lifetime utility over the three periods by choosing optimal borrowing and repayment strategies:

$$\max_{\{b_1, b_2, u_1\}} U(c_1) + \beta U(c_2) + \beta^2 U(c_3)$$

Where $U(\cdot)$ is an increasing concave utility function over consumption.

To solve for the optimal levels of borrowing (b_1^*, b_2^*) and unpaid debt (u_1^*), we derive the first-order conditions from the agent's intertemporal optimization problem. These take the form of Euler equations that characterize the marginal trade-offs across periods:

$$(b_1) \quad U'(c_1) = \beta U'(c_2) \left[(1 + r_1(b_1, k_1)) + b_1 \frac{\partial r_1}{\partial b_1} \right], \quad (\text{A1.3})$$

$$(b_2) \quad U'(c_2) = \beta U'(c_3) \left[(1 + r_2(b_2, k_2)) + b_2 \frac{\partial r_2}{\partial b_2} \right], \quad (\text{A1.4})$$

$$(u_1) \quad U'(c_2) = \beta U'(c_3) \left[(1 + \gamma) + b_2 \frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} \right]. \quad (\text{A1.5})$$

Equation (A1.3) governs the optimal borrowing decisions in period 1, accounting for how borrowing affects interest rates through marginal price effects. Equation (A1.4) does the same for period 2. Equation (A1.5) governs the optimal amount of unpaid debt u_1^* , where γ is the penalty or effective cost associated with carrying unpaid debt, and where the indirect effects of default on future borrowing costs are captured through the borrower-lender relationship channel (k_2).

A1.2. Initial Relationship and Individual's Default Behavior

We aim to understand how individuals adjust their repayment behavior in response to different levels of borrower-lender relationship. Specifically, we examine how the optimal level of unpaid debt, u_1^* , varies with the initial relationship level, $\bar{k}$. A higher initial value of $\bar{k}$ strengthens the borrower-lender relationship in subsequent periods for any given level of u_1 , which leads to a reduction in interest rates for any given borrowing amount.³² This, in turn, incentivizes individuals to borrow more which implies that: $\partial b_t^* / \partial \bar{k} \geq 0$.

³² For instance, holding all else constant, a borrower with higher initial relationship capital $\bar{k}$ can secure a larger loan in periods 1 and 2 at interest rates comparable to those offered to a borrower with lower $\bar{k}$.

Therefore, an increase in $\bar{k}$ has an unambiguously non-negative effect on the optimal borrowing levels (b_1^* and b_2^*). However, the effect on the optimal level of unpaid debt u_1^* is ambiguous. While a stronger initial relationship lowers borrowing costs and may reduce the incentive to default, it also encourages greater borrowing, which could increase the temptation or need to partially default. Therefore, the net effect on u_1^* is theoretically indeterminate. Given our interest in understanding the behavior of u_1^* , we focus on the case where $b_1^* > 0$, ensuring that borrowing is non-trivial and the trade-offs between borrowing and repayment are relevant.

Using Equations (A1.4) and (A1.5), and equating them, we obtain:

$$\left(1 + r_2(b_2^*, k_2)\right) + b_2^* \frac{\partial r_2}{\partial b_2} = (1 + \gamma) + b_2^* \frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1}$$

After the optimization process, the optimal values (b_1^* , b_2^* , u_1^*) can be expressed as functions of the initial relationship level, $\bar{k}$. To analyze how changes in $\bar{k}$ affect u_1^* , we differentiate both sides of the equation with respect to $\bar{k}$:

$$\frac{\partial}{\partial \bar{k}} \left[\left(1 + r_2(b_2^*, k_2)\right) + b_2^* \frac{\partial r_2}{\partial b_2} \right] = \frac{\partial}{\partial \bar{k}} \left[(1 + \gamma) + b_2^* \frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} \right]$$

Differentiating both sides with respect to $\bar{k}$ yields:

$$\begin{aligned} & \frac{\partial}{\partial \bar{k}} [r_2(b_2^*, k_2)] + \frac{\partial}{\partial \bar{k}} \left[b_2^* \frac{\partial r_2}{\partial b_2} \right] = \frac{\partial}{\partial \bar{k}} \left[b_2^* \frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} \right] \\ \rightarrow & \frac{\partial r_2}{\partial b_2} \frac{\partial b_2^*}{\partial \bar{k}} + \frac{\partial r_2}{\partial k_2} \left[\frac{\partial k_2}{\partial k_1} \frac{\partial k_1}{\partial \bar{k}} + \frac{\partial k_2}{\partial u_1} \frac{\partial u_1^*}{\partial \bar{k}} \right] + b_2^* \frac{\partial^2 r_2}{\partial b_2^2} \frac{\partial b_2^*}{\partial \bar{k}} + b_2^* \frac{\partial^2 r_2}{\partial b_2 \partial k_2} \left[\frac{\partial k_2}{\partial k_1} \frac{\partial k_1}{\partial \bar{k}} + \frac{\partial k_2}{\partial u_1} \frac{\partial u_1^*}{\partial \bar{k}} \right] = \\ & \frac{\partial b_2^*}{\partial \bar{k}} \frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} + b_2^* \frac{\partial k_2}{\partial u_1} \left[\frac{\partial^2 r_2}{\partial b_2 \partial k_2} \frac{\partial b_2^*}{\partial \bar{k}} + \frac{\partial^2 r_2}{\partial k_2^2} \left(\frac{\partial k_2}{\partial k_1} \frac{\partial k_1}{\partial \bar{k}} + \frac{\partial k_2}{\partial u_1} \frac{\partial u_1^*}{\partial \bar{k}} \right) \right] + b_2^* \frac{\partial r_2}{\partial k_2} \left[\frac{\partial^2 k_2}{\partial u_1^2} \frac{\partial u_1^*}{\partial \bar{k}} + \frac{\partial^2 k_2}{\partial u_1 \partial k_1} \frac{\partial k_1}{\partial \bar{k}} \right] \end{aligned}$$

Given that $k_1 = \bar{k}$, it follows that $\partial k_1 / \partial \bar{k} = 1$. Substituting this result gives:

$$\begin{aligned} & \frac{\partial u_1^*}{\partial \bar{k}} \left[b_2^* \frac{\partial^2 r_2}{\partial k_2^2} \left(\frac{\partial k_2}{\partial u_1} \right)^2 + b_2^* \frac{\partial r_2}{\partial k_2} \frac{\partial^2 k_2}{\partial u_1^2} - \frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} - b_2^* \frac{\partial^2 r_2}{\partial b_2 \partial k_2} \frac{\partial k_2}{\partial u_1} \right] = \\ & \left[\frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial k_1} + b_2^* \frac{\partial^2 r_2}{\partial b_2 \partial k_2} \frac{\partial k_2}{\partial k_1} - b_2^* \frac{\partial^2 r_2}{\partial k_2^2} \frac{\partial k_2}{\partial u_1} \frac{\partial k_2}{\partial k_1} - b_2^* \frac{\partial r_2}{\partial k_2} \frac{\partial^2 k_2}{\partial u_1 \partial k_1} \right] - \\ & \frac{\partial b_2^*}{\partial \bar{k}} \left[\frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} + b_2^* \frac{\partial^2 r_2}{\partial b_2 \partial k_2} \frac{\partial k_2}{\partial u_1} - 2 \frac{\partial r_2}{\partial b_2} - b_2^* \frac{\partial^2 r_2}{\partial b_2^2} \right] \end{aligned}$$

After rearranging the expression, we obtain:

$$\frac{\partial u_1^*}{\partial \bar{k}} = \underbrace{\left[\frac{\frac{\partial r_2}{\partial k_2} + b_2^* \frac{\partial^2 r_2}{\partial b_2 \partial k_2} - b_2^* \frac{\partial^2 r_2}{\partial k_2^2} \frac{\partial k_2}{\partial u_1} - b_2^* \frac{\partial r_2}{\partial k_2} \frac{\partial^2 k_2}{\partial u_1 \partial k_1}}{b_2^* \frac{\partial^2 r_2}{\partial k_2^2} \left(\frac{\partial k_2}{\partial u_1} \right)^2 + b_2^* \frac{\partial r_2}{\partial k_2} \frac{\partial^2 k_2}{\partial u_1^2} - \frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} - b_2^* \frac{\partial^2 r_2}{\partial b_2 \partial k_2} \frac{\partial k_2}{\partial u_1}} \right]}_A - \underbrace{\left[\frac{\frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} + b_2^* \frac{\partial^2 r_2}{\partial b_2 \partial k_2} \frac{\partial k_2}{\partial u_1} - 2 \frac{\partial r_2}{\partial b_2} - b_2^* \frac{\partial^2 r_2}{\partial b_2^2}}{b_2^* \frac{\partial^2 r_2}{\partial k_2^2} \left(\frac{\partial k_2}{\partial u_1} \right)^2 + b_2^* \frac{\partial r_2}{\partial k_2} \frac{\partial^2 k_2}{\partial u_1^2} - \frac{\partial r_2}{\partial k_2} \frac{\partial k_2}{\partial u_1} - b_2^* \frac{\partial^2 r_2}{\partial b_2 \partial k_2} \frac{\partial k_2}{\partial u_1}} \right] \frac{\partial b_2^*}{\partial \bar{k}}}_B$$

To analyze how changes in $\bar{k}$ affect the optimal unpaid debt, u_1^* , we examine the sign of the derivative $\partial u_1^* / \partial \bar{k}$, which is given by:

$$\frac{\partial u_1^*}{\partial \bar{k}} = A - B \cdot \frac{\partial b_2^*}{\partial \bar{k}}$$

The first term, A, represents the *short-run incentive to default*, capturing the immediate gains or losses from defaulting that are independent of future relationship considerations. In contrast, the second term, $B \cdot \frac{\partial b_2^*}{\partial \bar{k}}$, reflects the *long-run incentive to preserve the borrower-lender relationship*, as it indicates how future borrowing behavior responds to changes in relationship strength. To be more specific about the sign of the derivative $\partial u_1^* / \partial \bar{k}$, we adopt particular functional forms for the relationship variable k_t and the interest rate r_t , consistent with the properties outlined in Equations (A1.1) and (A1.2). Accordingly, we define k_t as:

$$k_t = k_{t-1} e^{\lambda(\bar{u} - u_{t-1})}$$

Where the sensitivity of the relationship to repayment performance is governed by the parameter $\lambda > 0$. We also define r_t as:

$$r_t = \begin{cases} r_a + \eta \frac{b_t^2}{k_t} & \text{if } b_t > 0 \text{ where } \eta = \frac{\alpha}{\mu}, \\ r_a & \text{if } b_t \leq 0. \end{cases}$$

Substituting these into the above and simplifying gives:

$$\frac{\partial u_1^*}{\partial \bar{k}} = \underbrace{\frac{1}{\lambda \bar{k}}}_{\text{Short-run incentive to default}} - \underbrace{\frac{3\eta b_2^* (\lambda b_2^* - 2) \partial b_2^*}{\lambda k_2 (r_a - \gamma) \partial \bar{k}}}_{\text{Long-run incentive to preserve relationship}} \quad (\text{A1.6})$$

This expression captures how the optimal level of u_1^* changes with respect to the initial borrower-lender relationship $\bar{k}$, which is independent of the specific functional form of the utility function.

The first term, $1/\lambda\bar{k}$, reflects the borrower's immediate trade-off: consuming more now at the expense of defaulting on loans which in turn damages the borrower-lender relationship. A higher $\bar{k}$ or a larger λ makes this fraction smaller, indicating that stronger or more sensitive relationships reduce the short-run temptation to default. In other words, when $\bar{k}$ is low, this term grows, indicating that a weak relationship encourages the borrower to favor extra consumption in period 2 despite the hit to future credit. Thus, $1/\lambda\bar{k}$ succinctly quantifies how individuals weigh the *present benefits* of default against the *immediate repercussions* for their banking relationship.

The second term, $\frac{3\eta b_2^*(\lambda b_2^* - 2)}{\lambda k_2(r_a - \gamma)} \frac{\partial b_2^*}{\partial \bar{k}}$, encapsulates how the borrower's debt level (b_2^*), the relationship sensitivity parameter (λ), and the convexity of borrowing costs (η) interact. A stronger relationship lowers interest rates, encouraging the borrower to take on more debt (b_2^*). However, this also increases the potential future cost of default, since damaging k_2 would raise interest rates in subsequent periods.

Whether this term pushes u_1^* down depends on the sign of $\lambda b_2^* - 2$ and on the net penalty $r_a - \gamma$. For example, if $\lambda b_2^* < 2$ and $r_a < \gamma$,³³ the term becomes negative, meaning the borrowers is more inclined to avoid default in order to protect access to cheaper future borrowing. This mechanism highlights how borrowers weigh *long-term credit access* against *short-term gains*.

Equation (A1.6) also reveals that the effect of the relationship strength $\bar{k}$ on the optimal default decision u_1^* is concave. When $\bar{k}$ is low, the borrower perceives limited future consequences from weakening the relationship, making the immediate consumption benefit from default more appealing. As $\bar{k}$ increases, the borrower takes on more debt in period 1 due to the lower interest rate, which raises the level of unpaid obligations. Beyond a certain point, however, the future costs of default become substantial, and the damage to the relationship is perceived as severe. This strengthens the incentive to preserve the relationship and reduces the likelihood of default.

In essence, weaker relationships lead to higher u_1^* , driven by short-term consumption motives, while stronger relationships lead to lower u_1^* , driven by the desire to preserve favorable credit terms. This concave relationship illustrates how borrower-lender relationships dynamically governs both borrowing behavior and repayment discipline over time.

What Strengthens the Long-Run Incentives?

The incentive to preserve the borrower-lender relationship over the long run is reinforced under three key conditions:

1. *Ongoing Credit Needs* ($b_2^* > 0$):

³³ When $r_a > \gamma$, the individual prefers to increase unpaid debt rather than increase borrowing.

Borrowers who anticipate future borrowing internalize the potential cost of damaging their relationship capital, k_2 . In contrast, borrowers who do not expect to borrow again – or who place little weight on future utility (i.e., has a low discount factor β) – derive minimal benefit from maintaining the relationship. These individuals are more likely to prioritize short-run consumption over long-run relational benefits.

2. *High Relationship Sensitivity (λ):*

When the relationship is highly sensitive to repayment behavior, even modest defaults can sharply deteriorate the relationship. This results in substantial increases in future interest rates, thereby intensifying the borrower's incentive to maintain a strong relationship.

3. *High Sensitivity of Borrowing Costs to Relationship (μ):*

If borrowing costs react strongly to variations in relationship strength, then even small declines in k_t can translate into significantly higher interest rates. This increases the value of preserving the relationship. Although a higher μ reduces the numerator of the *Relationship Preservation Incentive* term in Equation (A1.6) by lowering η , it also affects the term $\partial b_2^*/\partial \bar{k}$, which itself depends on η . Therefore, the total effect of changes in η must account for both its direct impact on the incentive term and its indirect effect via borrowing behavior, b_2^* .

When these conditions are met, the derivative $\partial u_1^*/\partial \bar{k}$ is more likely to be negative — indicating that borrowers are increasingly willing to limit their default rate in order to preserve access to favorable credit in future periods.

A1.3. Numerical Visualization

We numerically solve the individual's optimization problem in our three-period model to visualize how the optimal level of unpaid debt u_1^* varies with the strength of the initial borrower-lender relationship $\bar{k}$, under different values of the key parameters β , λ , η , and γ .

While a rigorous approach would involve calibrating the model to empirical data, such calibration lies beyond the scope of this study due to the lack of suitable data. Instead, we use a range of plausible parameter values and compute the first-order conditions over a grid of $\bar{k}$.

These numerical exercises confirm the analytically derived concave relationship between u_1^* and $\bar{k}$. They help illustrate the conditions under which long-run incentives to preserve the borrower-lender relationship outweigh the short-run gains from default, even when parameters vary within arbitrary but reasonable bounds.

[Insert Figure A1.1 About Here]

Figure A1.1 depicts how u_1^* responds to changes in the parameters:

- Lower β : When individuals place less weight on the future, they are more inclined to default, leading to higher unpaid debt. This illustrates a weakening of concern for long-term borrowing costs.
- Higher μ (or lower η): Greater sensitivity of borrowing costs to relationship strength enhances incentives to avoid default, thereby reducing u_1^* .
- Higher γ : The penalty on unpaid debt discourages default by raising its immediate cost.
- Higher λ : When future relationship strength is more sensitive to present behavior, borrowers are deterred from defaulting due to its amplified long-term consequences.

Importantly, the qualitative shape of the relationship between u_1^* and $\bar{k}$ is preserved across different functional forms for k_t and r_t , provided certain structural assumptions hold:

$$\frac{\partial k_t}{\partial u_{t-1}} < 0, \quad \frac{\partial^2 k_t}{\partial u_{t-1}^2} > 0, \quad \frac{\partial r_t}{\partial k_t} < 0, \quad \frac{\partial^2 r_t}{\partial k_t^2} > 0$$

Figure A1.2 demonstrates how different functional forms of k_t and r_t affect the curvature of the $\partial u_1^* / \partial \bar{k}$ relationship. The following forms are used:

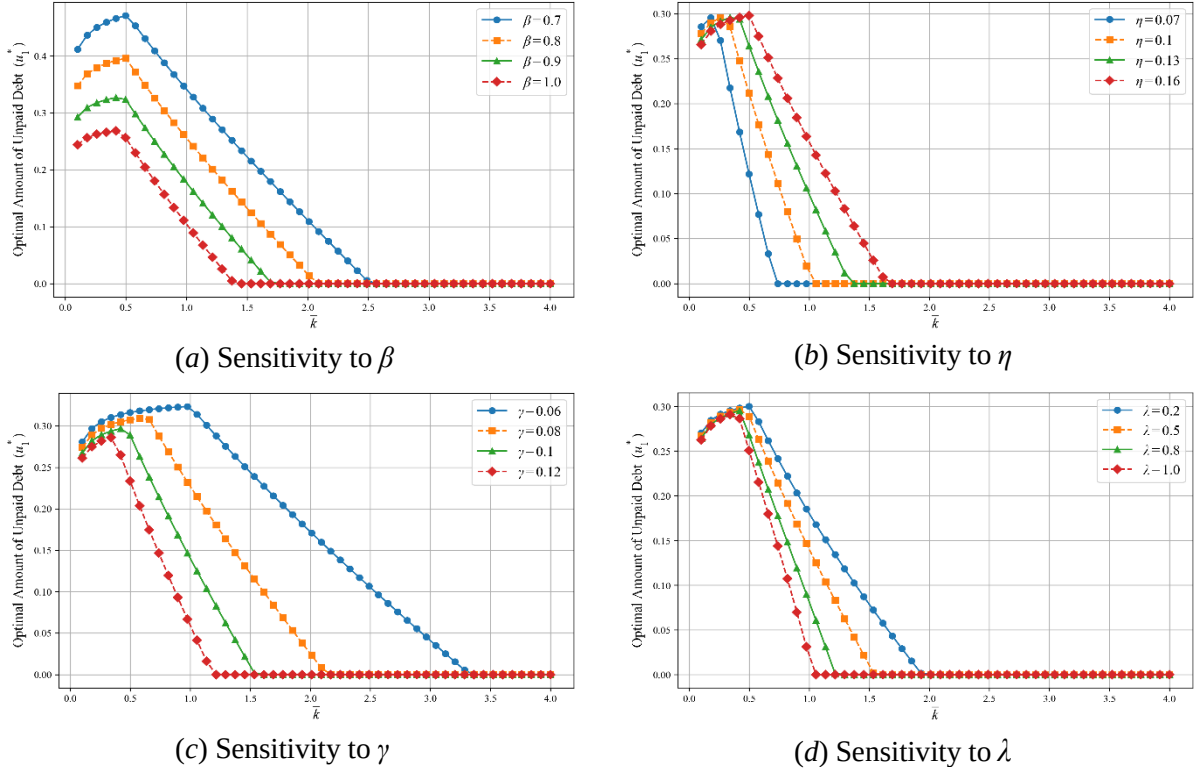
$$\begin{aligned} r_t^j &= r_a + \eta b_t^2 e^{-\mu k_t} && \text{for } b_t > 0, \\ r_t^j &= r_a + \frac{\eta \cdot \log(1 + b_t^2)}{k_t} && \text{for } b_t > 0, \\ k_t^j &= k_{t-1} (1 + \lambda(\bar{u} - u_{t-1})^\alpha) && \text{where } \alpha > 0, \\ k_t^j &= \frac{k_{t-1}^\alpha}{1 + \lambda(\bar{u} - u_{t-1})} && \text{where } \alpha > 1. \end{aligned}$$

Across these specifications, the results remain qualitatively consistent: a concave decline in u_1^* as $\bar{k}$ increases. However, the functional forms of k_t and r_t play a key role in shaping the borrower's short-run and long-run incentives to default. Notably, under the form $r_t = r_t^j$, the long-run incentive to preserve the borrower-lender relationship dominates even at relatively low levels of initial relationship strength, effectively suppressing the short-run incentive to default. Additionally, the functional form of $k_t = k_t^j$ makes the borrower less responsive to relationship preservation incentives, as this formulation is less sensitive to changes in the borrower's behavior.

[Insert Figure A1.2 About Here]

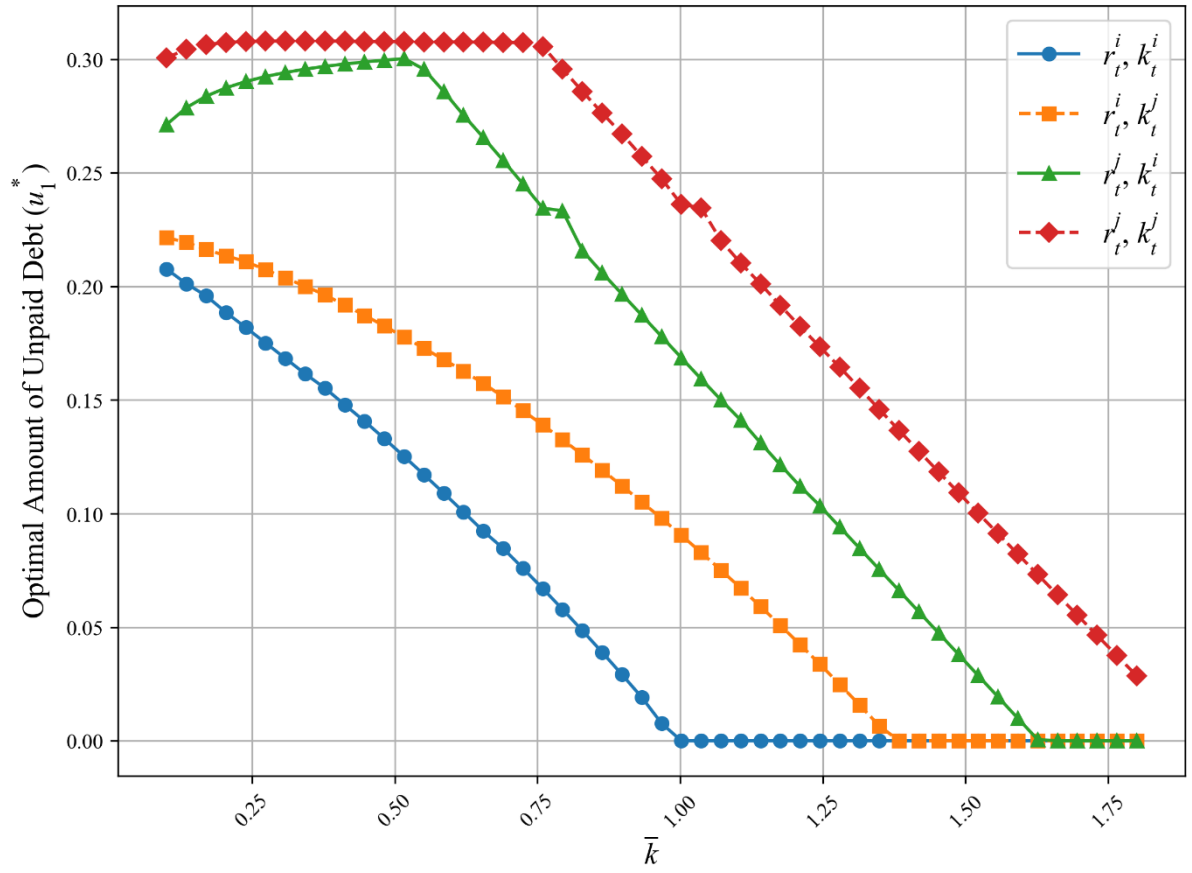
This theoretical model provides testable implications that guide our empirical investigation. It hypothesizes that borrowers with a stronger initial relationship with the bank will have lower incentives to default, as the future benefits of maintaining the relationship outweigh the short-term gains from shirking. In what follows, we empirically test this hypothesis using detailed data on mortgage loans and borrowers' characteristics, taking advantage of a unique institutional setting that allows us to distinguish between borrowers with and without prior relationships with the bank.

Figure A1.1: Sensitivity of the Optimal Amount of Unpaid Debt Level to Key Parameter Values



This figure illustrates how the optimal level of unpaid debt, u_1^* , varies with the strength of the borrower-lender relationship, $\bar{k}$, across different parameter values. Panels (a), (b), (c), and (d) show the results for variations in β , η , γ , and λ , respectively. In all panels, the x-axis represents the strength of the borrower-lender relationship, $\bar{k}$, while the y-axis shows the corresponding optimal level of unpaid debt, u_1^* .

Figure A1.2: Sensitivity of the Optimal Unpaid Debt Level to Alternative Functional Forms



This figure depicts how the optimal level of unpaid debt, u_1^* , changes with respect to the borrower-lender relationship strength, $\bar{k}$, under different functional forms for k_t and r_t . The x-axis represents the strength of the borrower-lender relationship, $\bar{k}$, while the y-axis indicates the corresponding optimal amount of unpaid debt, u_1^* .

Appendix A2. Comparison of Mortgage Loans Across the US, Europe, and Maskan Bank

Dimension	The U.S.	Europe	First Program	Second Program
Loan Structures	Predominantly 30-year fixed-rate mortgages (~ 3.5-5% during 2018-2020); 15-year fixed (~ 2.5-4.5% during 2018-2020); adjustable-rate mortgages (ARMs) also available; maximum LTV $\approx$ 97%; prepayment and refinancing generally allowed; standard amortization.	Mortgage terms vary by country, generally 15-30 years (some up to 40 years); fixed or floating rates (~ 1.5-2.5% between 2018-2020); LTV typically 80-90% (higher with guarantees); prepayment and refinancing generally allowed; standard amortization.	Only 12-year fixed-rate mortgage at 8%; ARMs also available; maximum LTV $\leq$ 80%; prepayment and refinancing generally allowed; standard amortization.	Only 12-year fixed-rate mortgage with nominal rate of 17.5% and effective rate is determined according to the market price of purchased mortgage rights (average effective rate $\sim$ 20.5%); maximum LTV $<$ 85% (average $\approx$ 35%); prepayment and refinancing generally allowed; standard amortization.
Requirements	Most borrowers require FICO $\geq$ 620 ($\geq$ 680 for highest LTV); income and steady employment verified; DTI $\leq$ 50%; FHA/VA/RHS programs available for low-income or first-time homebuyers (FICO $\geq$ 580).	Country-specific credit assessments; income continuity and savings verified; often requires 3+ years of salary documentation; credit registries (e.g., FICP, SCHUFA) track defaults; no unified EU-wide credit score; prime-income households favored; stricter rules for self-employed; DTI/debt-service ratios usually 30-40%.	Borrowers must have at least a one-year account in Maskan Bank; verified by Maskan Bank's internal credit evaluation and the central bank's assessment (based on records of bounced checks, loan defaults, tax arrears, unpaid utility bills, and average balances across all banks, not only Maskan Bank); holders of deposits with a minimum balance equal to at least 50% of the loan amount, maintained for at least one year; No maximum DTI ratio is required; Borrowers must be first-time homebuyers, and provide proof of marriage.	No need to have an account in Maskan Bank; verified only by central bank's assessment (based on records of bounced checks, loan defaults, tax arrears, unpaid utility bills, and average balances across all banks, not only Maskan Bank); Borrowers must acquire mortgage rights; No maximum DTI ratio is required; First-time homebuyer status is not required, nor is marriage.
Securitization	Deep secondary mortgage market; Fannie Mae and Freddie Mac issue government-backed MBS (~16% of mortgages by agencies, ~ 3% by private-label at end-2022); majority of loans securitized; extensive risk-sharing with investors; well-established ratings and insurance mechanisms.	Limited securitization; covered bonds (~ €500 billion in 2022) dominate; RMBS exist but are relatively small; most loans retained on bank balance sheets; modest use of STS framework; no broad secondary market comparable to the U.S.	Not securitized.	Not securitized.

This table presents a comparative overview of mortgage loan characteristics in the U.S., Europe, and Iran (Maskan Bank programs). Data sources include Freddie Mac's Primary Mortgage Market Survey (PMMS), historical mortgage data from the Federal Reserve Bank of St. Louis (FRED), the Fannie Mae Eligibility Matrix, the FDIC's Affordable Mortgage Lending Guide, Investopedia, the European Central Bank (ECB) and European Mortgage Federation (Hypostat Report), national credit registries such as FICP and SCHUFA, as well as official documents from Maskan Bank and the Central Bank of Iran.

Appendix A3. Variable Descriptions

Variable	Description
Bounced Check	A dummy variable that takes the value of one if a borrower has a history of bounced check with any bank in Iran.
Default in Previous Loans	A dummy variable that takes the value of one if a borrower has a default record in her history with any bank in Iran.
Distance to Maskan Branch (MBD)	It determines the borrower's residence 5-digit ZIP code center average distance to the k nearest branches, where k belongs to the set $\{1, 2, 3, \dots, 10\}$.
Distance Ratio (DR)	This variable introduces a distance ratio which is calculated due to dividing the borrower's residence 5-digit ZIP code center distance to the nearest Maskan Bank branch by the borrower's residence 5-digit ZIP code center average distance to the k nearest other banks branches, where k belongs to the set $\{1, 2, 3, \dots, 10\}$.
Education	A categorical variable which specifies the borrower's education level. The levels of education are categorized into major groups such as elementary school, high school, bachelor's degree, master's degree, and the PhD.
Effective Rate	The effective rate of a mortgage loan. It consists of the nominal interest rate of the loan and average market price of purchased mortgage right.
Home Price	The average home price per square meter in the borrower's neighborhood, calculated with the accuracy of the 5-digit Zip code. This data is reported monthly by the Ministry of Roads and Urban Development.
Job	A categorical variable which specifies the borrower's job category. The jobs in our data are categorized into major groups such as engineer, worker, manager, farmer, etc.
Loan to Value	The ratio of a loan to the value of its collateral.
Loan Origination Time	The number of days from the submission of mortgage application at a branch to loan approval.
Male	A dummy variable that takes the value of one if a borrower is male.
Married	A dummy variable that takes the value of one if a borrower is married.

Number of Maskan Branches (NMB)	This variable represents the number of Maskan Bank branches within a circle centered on the borrower's residence 5-digit ZIP code center, with radii of 500, 1000, 1500, and 2000 meters.
Quake	This variable denotes the treatment dummy which equals one if the borrower resides in the earthquake treated counties and zero otherwise.
RelationshipDummy	A dummy variable that takes the value of one if a borrower has a savings or current account prior to mortgage origination date.
RelationshipDuration	The number of years having a savings or current account prior to mortgage origination date.
3-Month (Weak) Default	A dummy variable equal to one if the borrower defaults within the full observation period after loan origination where the default is defined as failure to repay the installments for three consecutive months.
6-Month (Hard) Default	A dummy variable equal to one if the borrower defaults within the full observation period after loan origination where the default is defined as failure to repay the installments for six consecutive months.
6-Month Horizon (Weak) Default	A dummy variable that takes the value of one if the borrower defaults within the first 6 months after loan origination, and zero otherwise, where the default is defined as failure to repay the installments for three consecutive months.
12-Month Horizon (Weak) Default	A dummy variable that takes the value of one if the borrower defaults within the first 12 months after loan origination, and zero otherwise, where the default is defined as failure to repay the installments for three consecutive months.

This table presents the description of the variables used in this study.

Swiss Finance Institute

Swiss Finance Institute (SFI) is the national center for fundamental research, doctoral training, knowledge exchange, and continuing education in the fields of banking and finance. SFI's mission is to grow knowledge capital for the Swiss financial marketplace. Created in 2006 as a public-private partnership, SFI is a common initiative of the Swiss finance industry, leading Swiss universities, and the Swiss Confederation.